Fabius Integration Research Frontiers

A mixed-status source notebook for inversion, probability bridges, discrete engines, and saddle algebra

A research companion to *The Fabius Function and Rvachev's Up-Function*
based on Vladimir Reshetnikov's `ProveIt` formal development

Repository snapshot: 25 August 2026

Abstract

The long article *The Fabius Function and Rvachev's Up-Function* gives a broad human-readable account of the formal development in `ProveIt`. Its breadth is also its unavoidable weakness: several new branches of the Lean library postdate the article, and several generic proof engines are named only in the module guide because they have no counterpart in the exposition. This companion supplies those missing parts.

The first part develops the total inverse of the bounded Fabius function, transports infinite flatness into infinite endpoint steepness, and proves the strong non-elementarity results: no elementary function can agree with the Fabius function, Rvachev's up-function, or the signed global extension on any set with nonempty interior in the relevant support. The second part fills in the measure-theoretic bridges between the product probability law, its survival function, tilted Laplace moments, finite atomic laws, and half-weighted step approximants. The third part presents two reusable discrete engines omitted from the main article: weighted path sums for triangular recurrences and a finite-row Toeplitz convergence theorem. The opening audit also records a newly formalized redundancy in the original compact-support characterization: positivity of its dilation parameter follows from the remaining hypotheses. The final and largest part exposes the formal and analytic machinery behind the all-orders saddle expansion: parameter-dependent Poincaré expansions, recursive exponential and logarithmic coefficients, exact finite polynomial remainders, Gaussian contraction, uniform polynomial tails, the closed Stirling description of the saddle jets, and the explicit second periodic correction.

The purpose is complementarity rather than repetition. Definitions and headline theorems already proved in detail in the main article are recalled only when they are needed as input. Every substantial assertion below is either a direct human-readable rendering of a current Lean declaration, a proof-level explanation of a generic module that the main article explicitly omits, or a clearly identified elementary corollary of those declarations.

Formalization boundary. This is a research-frontier document. It preserves the source draft used during integration, including exact Lean-backed results, proof sketches, ordinary-analysis corollaries, stronger generalizations, and formulas that still lack exact declarations. Its presence under `non-formalized-research-frontiers` is intentional: no mathematical assertion in this notebook should be attributed to the formal library merely because a related definition or weaker theorem exists. The primary exposition contains the audited, declaration-backed subset. Among the obligations retained here are inverse curvature and Hölder claims, generic moment-matrix consequences, arbitrary-family Toeplitz wording, generic Gaussian coefficient bounds, and the passage from the mass top jet to the logarithmic highest-derivative law.

Contents

1 Scope, conventions, and the audit boundary 5
1.1 Three levels of coverage 5
1.2 Notation that must not collide 5
1.3 The positivity hypothesis in the original characterization is redundant 6

I The inverse and the elementary-function obstruction 6

2 The total inverse of the bounded Fabius function 6
2.1 Order-isomorphism construction and totalization 7
2.2 Reflection and exact inverse values 7
2.3 Formal interior calculus and an ordinary-analysis curvature corollary 8

3 Flatness of F becomes steepness of I 9
3.1 Root-free inverse bounds 9
3.2 An effective infinite-slope estimate 9

4 Elementary functions through their analytic locus 10
4.1 The exact formal class 10
4.2 Analytic loci 11
4.3 The reusable obstruction 12

5 Strong non-elementarity of the Fabius family 12

II Probability laws and the bridges the headline formulas conceal 13

6 The product probability law in measure-theoretic form 13
6.1 Sample space and uniform convergence 13
6.2 Deleting the head coordinate 14
6.3 Coordinate reflection 14

7 The survival-function Fubini identity 15

8 Raw, endpoint, and tilted Laplace moments 16
8.1 Three moment families 16
8.2 Differential recurrence 17
8.3 Reflection and the half moments 17

9 From atomic laws to half-endpoint histograms 17
9.1 Half-endpoint indicators are ordinary indicators almost everywhere 18
9.2 Atomic measure, cell geometry, and exact normalization 18
9.3 The overlap sandwich 19
9.4 Portmanteau plus a moving-boundary squeeze 19

III The reusable discrete engines 20

10 Weighted paths solve triangular recurrences **20**
10.1 Compositions as increasing paths . . . 20
10.2 Last-edge decomposition . . . 21
10.3 The generic triangular solver . . . 21

11 The nonrecursive composition formula for $F(2^{-n})$ **22**
11.1 Small cases as structural checks . . . 22

12 A finite-row Toeplitz convergence theorem **23**
12.1 Abstract statement . . . 23

13 The Fabius Toeplitz rows and corrected discrete prefixes **24**
13.1 The half-base q-binomial weights . . . 24
13.2 Why the inner range uses a half-cell correction . . . 25
13.3 Where the abstract theorem enters the discrete limit . . . 25

IV The formal and analytic machinery of the all-orders saddle expansion **25**

14 Parameter-dependent Poincaré expansions **26**
14.1 Elementary algebra of full expansions . . . 26
14.2 Flat functions . . . 27
14.3 Exact logarithmic coordinates . . . 27

15 Recursive coefficients of a formal exponential **27**
15.1 Structural laws . . . 28

16 Recursive coefficients of a formal logarithm **29**
16.1 From mass expansions to logarithmic expansions . . . 30

17 Exact finite remainders from formal coefficient identities **30**

18 Gaussian contraction: integration becomes polynomial algebra **31**
18.1 Gaussian moments . . . 31
18.2 The contraction functional . . . 32

19 Uniform Gaussian tails for polynomial references **32**
19.1 A factorial coefficient weight . . . 32
19.2 The all-orders central radius . . . 33

20 Closed saddle jets and shifted Stirling numbers **33**
20.1 The exponent coefficient at arbitrary order . . . 34

21 From exponent coefficients to periodic mass and log coefficients **35**
21.1 The highest derivative has a universal coefficient . . . 35

22 The second periodic correction, line by line **36**
22.1 Gaussian contraction . . . 36
22.2 Taking the logarithm . . . 37

23 Assembly of the full endpoint expansion **37**
23.1 Exact Lambert phase and exact normalized mass 37
23.2 The central integral and the moving window 38
23.3 Finite exponential substitution 39
23.4 Gaussian contraction and the disappearance of half-orders 39
23.5 Logarithmic transfer and restoration of the flat tail 39

24 Why the exact phase cannot be casually re-expanded **40**

A Executable recurrences in mathematical pseudocode **40**

B Lean declaration and module concordance **42**

C Audit matrix against the main article **44**

D Notation crosswalk **45**

E A compact dependency diagram **46**

1 Scope, conventions, and the audit boundary

Let $F : \mathbb{R} \to [0,1]$ denote the bounded Fabius function, totalized in the usual way by $F(x) = 0$ for $x \le 0$ and $F(x) = 1$ for $x \ge 1$. Let

$$\mathrm{up}(x) = F(1 - |x|) \tag{1}$$

be Rvachev's compactly supported up-function, and let $\widetilde{F}$ denote the signed global extension used in the Lean development. On $[0,1]$ the basic identities are

$$F(1-x) = 1 - F(x), \qquad F'(x) = 2\,\mathrm{up}(2x-1), \qquad F'(x) = 2F(2x) \quad (0 < x < \tfrac{1}{2}). \tag{2}$$

These are inputs here, not topics to be reproved.

The repository snapshot underlying this supplement is the head of `main` on 25 August 2026, commit

`5e80805b1ee92697723949d125aa0d6dbf32f538`.

The main article itself records an earlier pinned snapshot and, near the beginning of its module guide, explicitly says that four clusters have "no counterpart anywhere in the article": Gaussian contraction and Gaussian tails, the formal saddle exponential and logarithm algebra, the abstract Toeplitz row lemma, and the generic weighted-path solver. Those four clusters receive complete mathematical chapters below. Two additional clusters, the inverse and non-elementarity developments, are absent because they were added after the article's pinned snapshot. The current snapshot also adds a proof that positivity of the dilation parameter in the original compact-support characterization is redundant. Finally, several bridges are present only as one- or two-sentence transitions; they are expanded here into proof-level arguments.

1.1 Three levels of coverage

To keep the audit precise, each topic belongs to one of the following levels.

Status	Meaning in this companion
New branch	A current Lean module whose mathematical content does not occur in the main article: principally `FabiusInverse.lean`, `ElementaryFunction.lean`, `NotElementary.lean`, and `ProbabilityLaplaceMoments.lean`.
Missing engine	A generic theorem family that the main article deliberately leaves in the module guide: path sums, Toeplitz rows, formal coefficient recurrences, finite remainders, and Gaussian polynomial estimates.
Compressed bridge	A statement whose endpoints occur in the main article but whose intermediate measure theory, topology, or asymptotic algebra is suppressed.

1.2 Notation that must not collide

The exact half moments are denoted by d_n, following the main article. The bounded saddle exponent jets are instead denoted by

$$Z_n(t), \tag{3}$$

although their Lean declaration is `negativeLaplaceBoundedExponentJet`. This avoids using the same letter for two wholly different sequences. We write $L = \log 2$, Ψ for the smooth one-periodic correction in the negative-Laplace exponent, and $\lambda(x)$ for the exact lower-Lambert phase at small positive x.

Formal boundary convention. When a statement is a theorem in the current Lean library, the corresponding declaration is named in the margin of the discussion or in the concordance in section B. When a statement is an immediate classical calculus corollary but is not presently packaged as a declaration, it is called a *corollary in ordinary analysis.* This distinction matters most for the second-derivative curvature formula of the inverse and for the transfer of the highest-jet coefficient from mass coefficients to logarithmic coefficients.

1.3 The positivity hypothesis in the original characterization is redundant

The compact-support characterization used in the first Arias de Reyna paper [3] assumes a smooth function $\phi : \mathbb{R} \to \mathbb{R}$, a real dilation parameter k, and

$$\operatorname{tsupp} \phi = [-1, 1], \qquad \phi(x) > 0 \quad (-1 < x < 1), \qquad \phi(0) = 1, \tag{4}$$

together with the derivative law

$$\phi'(x) = k\big(\phi(2x+1) - \phi(2x-1)\big). \tag{5}$$

The paper states $k > 0$ as a hypothesis. In the current formal development that field is retained for source fidelity, but a smart constructor proves that its positivity follows from the other assumptions.

Theorem 1.1 (Redundancy of scale positivity)**.** *Suppose $\phi \in C^\infty(\mathbb{R})$ satisfies (4) and (5) for some $k \in \mathbb{R}$. Then $k > 0$. Consequently these data determine an instance of the full original-characterization predicate without a separate positivity proof.*

Proof. The support identity and continuity imply $\phi(-1) = 0$: the function vanishes on $(-\infty, -1)$, and -1 is a limit point of that interval. The mean-value theorem on $[-1, 0]$ therefore gives a point $c \in (-1, 0)$ such that

$$\phi'(c) = \frac{\phi(0) - \phi(-1)}{0 - (-1)} = 1. \tag{6}$$

Now $2c - 1 < -1$, so the support condition gives $\phi(2c - 1) = 0$, whereas $-1 < 2c + 1 < 1$, so interior positivity gives $\phi(2c + 1) > 0$. Evaluating (5) at c yields

$$1 = k\phi(2c + 1). \tag{7}$$

The second factor is positive, hence $k > 0$. □

This is `IsOriginalFabius.mk_of_derivative_law`. The proof is small, but it clarifies the logical content of the original theorem: the sign of the scale is forced before the later Fourier argument identifies its exact value $k = 2$.

Part I
The inverse and the elementary-function obstruction

2 The total inverse of the bounded Fabius function

The main article proves that F is a continuous strictly increasing bijection of $[0, 1]$ onto itself. The inverse development turns that qualitative fact into a reusable global object whose endpoint behavior can be estimated exactly.

2.1 Order-isomorphism construction and totalization

Let $F_{[0,1]}$ be the restriction of F to the subtype $[0,1]$. Strict monotonicity and surjectivity package it as an order isomorphism

$$\mathfrak{F} : [0,1] \simeq_o [0,1]. \tag{8}$$

Define the clamping map

$$\pi(y) = \min(1, \max(0, y)) \tag{9}$$

and the total inverse

$$I(y) = \operatorname{val}\big(\mathfrak{F}^{-1}(\pi(y))\big). \tag{10}$$

This is the mathematical content of `fabiusOrderIso` and `fabiusInv`; Lean uses `Set.IccExtend` rather than spelling out (9).

Theorem 2.1 (Global inverse package)**.** *The function $I : \mathbb{R} \to \mathbb{R}$ has the following properties.*

1. $0 \le I(y) \le 1$ *for every* $y \in \mathbb{R}$.
2. I *is continuous and nondecreasing on* $\mathbb{R}$.
3. I *is strictly increasing on* $[0,1]$ *and maps that interval bijectively onto itself.*
4. *For* $x, y \in [0,1]$,
$$F(I(y)) = y, \qquad I(F(x)) = x. \tag{11}$$
5. *The clamped tails are*
$$I(y) = 0 \quad (y \le 0), \qquad I(y) = 1 \quad (y \ge 1). \tag{12}$$

Proof. The inverse of an order isomorphism is again an order isomorphism, hence continuous for the order topologies and strictly increasing. The interval projection π is continuous and nondecreasing, so the composite (10) is continuous and nondecreasing on all of $\mathbb{R}$. Its values lie in $[0,1]$ because the codomain of $\mathfrak{F}^{-1}$ is that subtype. On $[0,1]$ the projection is the identity, and the two inverse identities are the usual *apply–inverse–apply* laws for $\mathfrak{F}$. If $y \le 0$, then $\pi(y) = 0$ and $\mathfrak{F}^{-1}(0) = 0$ because $F(0) = 0$; the right tail is identical. □

The comparison laws are often more useful than direct substitution.

Proposition 2.2 (Comparison calculus)**.** *For* $x, y \in [0,1]$,

$$I(y) \le x \iff y \le F(x), \qquad I(y) < x \iff y < F(x), \tag{13}$$
$$x \le I(y) \iff F(x) \le y, \qquad x < I(y) \iff F(x) < y. \tag{14}$$

Proof. Apply the strictly increasing map F to the first comparison, or the strictly increasing map I to the second, and use (11). The strict versions are the same argument with $<$ in place of $\le$. □

2.2 Reflection and exact inverse values

The reflection symmetry of F survives totalization without a domain restriction.

Theorem 2.3 (Global reflection)**.** *For every* $y \in \mathbb{R}$,

$$I(1 - y) = 1 - I(y). \tag{15}$$

In particular

$$I(0) = 0, \qquad I(1) = 1, \qquad I\left(\frac{1}{2}\right) = \frac{1}{2}. \tag{16}$$

Proof. If $y \le 0$ or $y \ge 1$, the identity follows directly from the two constant tails. Suppose $0 < y < 1$. Both $I(1-y)$ and $1 - I(y)$ lie in $[0,1]$. Applying F and using the reflection law for F gives

$$F(1 - I(y)) = 1 - F(I(y)) = 1 - y = F(I(1-y)).$$

Injectivity of F on $[0,1]$ proves (15). Substituting $y = 1/2$ then forces the midpoint identity. □

Exact dyadic evaluation of F immediately becomes exact evaluation of I at a dense family of rational ordinates. Let $D_{n,a}$ denote the exact bounded dyadic evaluator, so that for $0 \le a \le 2^n$,

$$D_{n,a} = F\left(\frac{a}{2^n}\right) \in \mathbb{Q}. \tag{17}$$

The evaluator is clamped for $a > 2^n$.

Theorem 2.4 (Inverting the exact dyadic evaluator)**.** *For all* $n, a \in \mathbb{N}$,

$$I(D_{n,a}) = \frac{\min(a, 2^n)}{2^n}. \tag{18}$$

In particular,

$$F\left(\frac{1}{4}\right) = \frac{5}{72} \quad \Longrightarrow \quad I\left(\frac{5}{72}\right) = \frac{1}{4}. \tag{19}$$

Proof. For $a \le 2^n$, combine (17) with the left inverse identity in (11). For $a \ge 2^n$, the clamped evaluator equals 1 and the result is $I(1) = 1$. □

2.3 Formal interior calculus and an ordinary-analysis curvature corollary

The inverse is constructed using order topology alone. Once the positivity theorem for F' in the open unit interval is imported, the local inverse theorem supplies a sharper interior picture; the formal development now packages both the reciprocal derivative and full interior smoothness directly.

Corollary 2.5 (Interior derivative of the inverse)**.** *For* $0 < y < 1$,

$$I'(y) = \frac{1}{F'(I(y))} = \frac{1}{2\,\mathrm{up}(2I(y) - 1)}. \tag{20}$$

Thus I *is* C^∞ *on* $(0,1)$.

Proof. The point $x = I(y)$ lies in $(0,1)$, and strict positivity of $F'(x)$ there makes F a local C^∞ diffeomorphism. Differentiate $F(I(y)) = y$ and substitute the second identity in (2). □

Because F is convex on $(0, 1/2)$ and concave on $(1/2, 1)$, inversion reverses those curvature signs.

Corollary 2.6 (Curvature of the inverse)**.** *The inverse* I *is concave on* $(0,1/2)$ *and convex on* $(1/2,1)$. *Where second derivatives are evaluated,*

$$I''(y) = -\frac{F''(I(y))}{F'(I(y))^3}. \tag{21}$$

The midpoint is the reflection-symmetric change of curvature.

Remark 2.7. The first identity in theorem 2.5, its explicit up-form, strict positivity, and the resulting interior smoothness are the named Lean declarations `fabiusInv_hasDerivAt`, `deriv_fabiusInv`, `deriv_fabiusInv_eq_inv_two_mul_rvachevUp`, and `deriv_fabiusInv_pos`, together with `fabiusInv_contDiffOn_Ioo`. The second-derivative curvature formula in theorem 2.6 remains an ordinary-analysis corollary rather than a separate declaration.

3 Flatness of F becomes steepness of I

The inverse is most interesting at the two endpoints, where the classical formula (20) ceases to have a finite limit. The formal development proves this without differentiating the inverse: exact upper bounds for F are transported through monotonicity and then solved as inequalities for I.

3.1 Root-free inverse bounds

For $n \in \mathbb{N}$, the effective flatness theorem for F says that on the dyadic window $0 \le x \le 2^{-n}$,

$$F(x) \le 2^{\binom{n+1}{2}} x^n. \tag{22}$$

The sharp factorial refinement is

$$F(x) \le \frac{2^{\binom{n+1}{2}}}{n!} x^n. \tag{23}$$

Substitute $x = I(y)$ and use $F(I(y)) = y$.

Theorem 3.1 (Flatness transported through the inverse). *Let $n \in \mathbb{N}$ and $y \in [0,1]$. If*

$$2^n I(y) \le 1, \tag{24}$$

then

$$y \le 2^{\binom{n+1}{2}} I(y)^n, \tag{25}$$

$$y \le \frac{2^{\binom{n+1}{2}}}{n!} I(y)^n. \tag{26}$$

It is enough to assume the argument-side condition

$$0 \le y \le F(2^{-n}). \tag{27}$$

Proof. Under (24), the point $x = I(y)$ belongs to the window of (22) and (23). Their left sides become $F(I(y)) = y$, giving the two conclusions. Under (27), the comparison calculus (13) gives $I(y) \le 2^{-n}$, which is exactly (24). □

If roots are allowed, (26) reads

$$I(y) \ge \left(\frac{n!}{2^{\binom{n+1}{2}}} y\right)^{1/n} \qquad \big(0 \le y \le F(2^{-n})\big). \tag{28}$$

The Lean statements retain the root-free form because all constants remain exact and no case split for real powers is required.

3.2 An effective infinite-slope estimate

The case $n = 2$ already forces the difference quotient to diverge. Since $2^{\binom{3}{2}} = 8$ and $F(1/4) = 5/72$, (25) gives

$$y \le 8I(y)^2 \qquad \left(0 \le y \le \frac{5}{72}\right). \tag{29}$$

Theorem 3.2 (Effective domination of every slope). *Let $M \in \mathbb{R}$ and $y > 0$. If*

$$y \le F(1/4) = \frac{5}{72}, \qquad 8M^2 y < 1, \tag{30}$$

then

$$My < I(y). \tag{31}$$

Proof. Assume contrariwise that $I(y) \leq My$. The left side is nonnegative, so squaring preserves the inequality and gives $I(y)^2 \leq M^2y^2$. Combining this with (29) yields

$$y \leq 8M^2y^2.$$

Since $y > 0$, division by y gives $1 \leq 8M^2y$, contradicting (30). □

No positivity assumption on M is necessary: for $M \leq 0$ the conclusion is automatic from $I(y) > 0$, and the same algebra proves it uniformly.

Theorem 3.3 (Infinite one-sided endpoint slopes)**.** *The endpoint difference quotients diverge:*

$$\frac{I(y)}{y} \longrightarrow +\infty \qquad (y \downarrow 0), \tag{32}$$

$$\frac{1 - I(y)}{1 - y} \longrightarrow +\infty \qquad (y \uparrow 1). \tag{33}$$

Proof. Fix a target slope M. Replace it by $|M| + 1$ and choose $y > 0$ below both $F(1/4)$ and $[8(|M| + 1)^2]^{-1}$. Then theorem 3.2 gives $(|M| + 1)y < I(y)$, hence $M < I(y)/y$. This is exactly convergence to $+\infty$ in the order topology. For the right endpoint, put $z = 1 - y$ and use $I(1 - z) = 1 - I(z)$. □

Corollary 3.4 (Failure of endpoint Lipschitz and Hölder bounds)**.** *The inverse is not locally Lipschitz at* 0 *or* 1*. More strongly, for every* $\alpha > 0$ *and every* $C > 0$*, no estimate*

$$I(y) \leq Cy^{\alpha} \tag{34}$$

can hold throughout a punctured right neighborhood of 0*; the reflected statement holds at* 1*.*

Proof. The Lipschitz claim follows immediately from (32). For the Hölder claim, choose n with $1/n < \alpha$. On the shrinking window $0 < y \leq F(2^{-n})$, (28) gives $I(y) \geq c_n y^{1/n}$ with $c_n > 0$. If (34) held, then $c_n \leq Cy^{\alpha - 1/n}$, whose right side tends to zero as $y \downarrow 0$, a contradiction. □

The windows $y \leq F(2^{-n})$ shrink super-exponentially as n grows. One must therefore not read (28) as a lower bound uniform in n. Its strength is quantifier order: for each prescribed root exponent one obtains a sufficiently small window. That is exactly what is needed to defeat every fixed positive Hölder exponent.

---

4 Elementary functions through their analytic locus

The non-elementarity proof does not attempt differential algebra or Liouville theory. It uses a topological-analytic invariant that is both stronger and easier to formalize: every one-variable elementary function is analytic on a dense open set, whereas the Fabius family is nonanalytic at every point of the relevant interval.

4.1 The exact formal class

Definition 4.1 (The class Elem)**.** Let Elem be the smallest class of total functions $\mathbb{R} \to \mathbb{R}$ containing all constant functions and the identity and closed under

$$+, \quad \cdot, \quad -, \quad (\cdot)^{-1}, \quad f \mapsto f^r \ (r \in \mathbb{R}), \quad \exp, \ \log, \ \sin, \ \cos, \ \arcsin, \ \arctan, \tag{35}$$

where every unary operation is applied pointwise.

This is the inductive predicate `IsElementary`. Composition is deliberately not a constructor.

Proposition 4.2 (Closure derived from the generators)**.** *The class* Elem *is closed under composition. It is consequently closed under subtraction, division, natural powers, polynomial and rational functions, square roots, absolute values, tangent, hyperbolic functions,* arccos, arsinh*, and signed odd roots.*

Proof. Induct on the construction of the outer function g. Precomposing a constant or the identity with an elementary f is elementary. Every remaining constructor commutes with precomposition: for example $(g_1 + g_2) \circ f = (g_1 \circ f) + (g_2 \circ f)$, and similarly for the unary generators. This proves closure under composition. The remaining operations are identities in the generated algebra. For example

$$|f| = \sqrt{f^2}, \qquad \tan f = \frac{\sin f}{\cos f}, \qquad \sinh f = \frac{\mathrm{e}^f - \mathrm{e}^{-f}}{2}.$$

For an odd positive integer n, the signed root may be written

$$\frac{f}{|f|}|f|^{1/n},$$

with value zero at $f = 0$ under the total reciprocal convention. □

Remark 4.3 (Totalization makes the theorem stronger). Mathlib treats reciprocal, logarithm, real power, and inverse trigonometric functions as total maps on $\mathbb{R}$; outside their classical domains they receive specified extension values. Thus theorem 4.1 is at least as large as the class represented by classical expressions on open domains. Proving that F does not belong to this larger class is a stronger statement, not a domain-convention loophole.

4.2 Analytic loci

For a function $f : \mathbb{R} \to \mathbb{R}$, define

$$\mathcal{A}(f) = \{x \in \mathbb{R} : f \text{ is real analytic at } x\}. \tag{36}$$

Analyticity is local, hence $\mathcal{A}(f)$ is open for every f. The substantive theorem is density for elementary functions.

The only difficult induction step is composition with a unary generator that has finitely many exceptional values. The following lemma isolates it.

Lemma 4.4 (Finite-bad-value composition principle)**.** *Let* $f, g : \mathbb{R} \to \mathbb{R}$*. Suppose that* $\mathcal{A}(f)$ *is dense and that there is a finite set* $B \subset \mathbb{R}$ *such that* g *is analytic at every point of* $\mathbb{R} \setminus B$*. Then* $\mathcal{A}(g \circ f)$ *is dense.*

Proof. Let U be a nonempty open interval. Because $\mathcal{A}(f)$ is dense and open, choose $x_0 \in U \cap \mathcal{A}(f)$. We prove that every punctured neighborhood of x_0 contains a point where $g \circ f$ is analytic.

Fix $c \in B$. The analytic function $f - c$ near x_0 has two possibilities.

1. It vanishes identically on a neighborhood of x_0. Then f is locally constant equal to c, so $g \circ f$ is locally constant and therefore analytic at x_0, regardless of whether g is analytic at c.
2. It is not locally zero. The isolated-zero theorem gives a punctured neighborhood of x_0 on which $f(x) \neq c$.

If the first alternative occurs for any $c \in B$, we are done. Otherwise intersect the finitely many punctured neighborhoods. The intersection still contains points arbitrarily close to x_0; intersect once more with the open set $U \cap \mathcal{A}(f)$. At any resulting point x, the value $f(x)$ avoids B, so g is analytic at $f(x)$ and f is analytic at x. The analytic chain rule makes $g \circ f$ analytic at x. Thus every nonempty open U meets $\mathcal{A}(g \circ f)$. □

Theorem 4.5 (Dense open analytic locus). *If $f \in \mathsf{Elem}$, then $\mathcal{A}(f)$ is dense and open. Equivalently, the nonanalytic locus*

$$\mathbb{R} \setminus \mathcal{A}(f) \tag{37}$$

is closed and nowhere dense.

Proof. Openness is true for every analytic locus. Density follows by induction on the elementary construction.

Constants and the identity are analytic everywhere. For sums and products, the intersection of the two dense open analytic loci is dense and each point in the intersection is analytic for the result. Negation is immediate. For a unary generator, apply theorem 4.4 with the following exceptional sets:

Generator	Exceptional values	Analytic elsewhere
$t \mapsto t^{-1}$	$\{0\}$	yes
$t \mapsto t^r$	$\{0\}$	yes for every fixed $r \in \mathbb{R}$
$\log t$	$\{0\}$	yes under the total real-log convention
$\arcsin t$	$\{-1, 1\}$	yes
$\exp t, \sin t, \cos t, \arctan t$	$\varnothing$	everywhere

The complement of a dense open set is closed with empty interior, hence nowhere dense. □

4.3 The reusable obstruction

Theorem 4.6 (No agreement on an interior). *Let $h : \mathbb{R} \to \mathbb{R}$ be nonanalytic at every point of $\operatorname{int}(S)$, where $S \subset \mathbb{R}$ has nonempty interior. No elementary function g can agree with h on all of S.*

Proof. If g were elementary, theorem 4.5 would give a point $x \in \operatorname{int}(S)$ at which g is analytic. Equality on S implies equality on a whole neighborhood of x contained in S, so analyticity transfers from g to h at x, a contradiction. □

The theorem is stronger than saying that h itself is not elementary: it rules out any piecewise repair outside S and any elementary surrogate on a substantial subset.

5 Strong non-elementarity of the Fabius family

The main article proves nowhere analyticity of F on $[0, 1]$, of up on $[-1, 1]$, and of the signed extension on its active interval. Combining those results with theorem 4.6 gives the new non-elementarity branch almost for free.

Theorem 5.1 (Strong local non-elementarity of F). *Let $U \subset [0, 1]$ have nonempty interior. There is no $g \in \mathsf{Elem}$ such that*

$$g(x) = F(x) \qquad (x \in U). \tag{38}$$

In particular, no elementary function agrees with F on $(0, 1)$, and F is not an elementary function on $\mathbb{R}$.

Proof. The function F is nonanalytic at every point of $[0, 1]$, hence at every point of $\operatorname{int}(U)$. Apply theorem 4.6. □

Theorem 5.2 (Strong local non-elementarity of up). *If $U \subset [-1, 1]$ has nonempty interior, no elementary g agrees with* up *on U. Consequently* up *is not elementary.*

Theorem 5.3 (Strong local non-elementarity of the signed extension). *If $U \subset [0, 2)$ has nonempty interior, no elementary g agrees with the signed global extension $\widetilde{F}$ on U. Consequently $\widetilde{F}$ is not elementary.*

Proof of the last two theorems. The corresponding nowhere-analyticity theorems supply the hypothesis of theorem 4.6. The global conclusions follow by taking a nonempty open subinterval of the active support and observing that an elementary representation on all of $\mathbb{R}$ would in particular agree there. □

Remark 5.4 (Why compact support alone is not the argument). A compactly supported nonzero function cannot be globally real analytic, but elementary functions need not be analytic everywhere: $|x|$, x^{-1}, and totalized logarithms already have exceptional points. What defeats elementarity is not compact support by itself; it is nonanalyticity on an entire interval, while elementary exceptional sets are nowhere dense.

Remark 5.5 (Why this is sharper than a differential-algebra slogan). The conclusion does not depend on choosing a particular syntax for "closed form" beyond the explicit inductive class. More importantly, it is local: changing the function outside a small open interval cannot make it elementary on that interval. Thus the theorem rules out not only a global elementary formula for F, but also an elementary formula valid on any nontrivial open patch of its natural domain.

Part II
Probability laws and the bridges the headline formulas conceal

6 The product probability law in measure-theoretic form

The random-series representation in the main article is concise because it speaks in the usual probabilistic shorthand. The formal construction must establish continuity, measurability, product independence, pushforward identities, and support before it may use the same shorthand. This section records that chain explicitly.

6.1 Sample space and uniform convergence

Let
$$\Omega = [0,1]^{\mathbb{N}} \tag{39}$$
with its product topology and product Borel probability measure
$$\mathbf{P} = \bigotimes_{n\geq 0} \mathrm{Unif}[0,1]. \tag{40}$$

Write ω_n for the nth coordinate and define
$$X(\omega) = \sum_{n=0}^{\infty} \frac{\omega_n}{2^{n+1}}. \tag{41}$$

For every ω,
$$0 \leq \frac{\omega_n}{2^{n+1}} \leq 2^{-n-1}, \qquad \sum_{n\geq 0} 2^{-n-1} = 1. \tag{42}$$

The Weierstrass test therefore gives uniform convergence on Ω. Every partial sum is continuous, so X is continuous and hence measurable. Termwise comparison in (42) gives the pointwise support statement
$$0 \leq X(\omega) \leq 1. \tag{43}$$

Let

$$\mu = \mathbf{P} \circ X^{-1} \tag{44}$$

be the pushforward law. It is a probability measure and

$$\mu([0,1]) = 1, \qquad \mu(\mathbb{R} \setminus [0,1]) = 0. \tag{45}$$

The Lean declarations are `weightedCoordinateSum`, `continuous_weightedCoordinateSum`, `weightedSumDistribution`, and the support lemmas surrounding them.

6.2 Deleting the head coordinate

Define the left shift

$$(T\omega)_n = \omega_{n+1}. \tag{46}$$

The product measure is invariant under this deletion in the sense that $T_{\#}\mathbf{P} = \mathbf{P}$, and the head coordinate ω_0 is independent of the entire tail $T\omega$. Reindexing the series gives the exact pointwise split

$$X(\omega) = \frac{\omega_0 + X(T\omega)}{2}. \tag{47}$$

Consequently the pair $(\omega_0, X(T\omega))$ has law

$$\mathrm{Unif}[0,1] \otimes \mu. \tag{48}$$

Pushing (47) through this joint law gives the self-similarity identity

$$\mu = (\mathrm{Unif}[0,1] \otimes \mu) \circ \left[(u,x) \mapsto \frac{u+x}{2}\right]^{-1}. \tag{49}$$

Proposition 6.1 (The smoothing equation for the CDF). *Let $H(y) = \mu((-\infty, y])$. Then for every $y \in \mathbb{R}$,*

$$H(y) = \int_0^1 H(2y-u)\,\mathrm{d}u. \tag{50}$$

For $0 \le y \le 1/2$ this reduces to

$$H(y) = \int_0^{2y} H(t)\,\mathrm{d}t. \tag{51}$$

Proof. Condition on the head coordinate in (47). For a fixed u,

$$\mathbb{P}\left(\frac{u+X'}{2} \le y\right) = \mathbb{P}(X' \le 2y-u) = H(2y-u),$$

where X' has law μ and is independent of u. Integrate over $u \in [0,1]$. If $0 \le y \le 1/2$, support implies $H(2y-u) = 0$ when $u > 2y$; substitute $t = 2y-u$ and use the reflection identity established below, or reverse the interval directly. □

6.3 Coordinate reflection

Let $R : \Omega \to \Omega$ be

$$(R\omega)_n = 1 - \omega_n. \tag{52}$$

Lebesgue measure on $[0,1]$ is invariant under $u \mapsto 1-u$, so the infinite product measure is invariant under R. Moreover

$$X(R\omega) = \sum_{n \ge 0} \frac{1-\omega_n}{2^{n+1}} = 1 - X(\omega). \tag{53}$$

Thus the law is reflection invariant:

$$(x \mapsto 1 - x)_{\#}\mu = \mu. \tag{54}$$

Since the CDF is continuous, this becomes

$$H(1 - y) = 1 - H(y). \tag{55}$$

The fixed-point uniqueness theorem for the bounded Fabius equation now identifies

$$H = F \quad \text{on all of } \mathbb{R}. \tag{56}$$

The phrase "on all of $\mathbb{R}$" includes the two constant tails; it is not a reference to the signed extension $\widetilde{F}$.

Theorem 6.2 (Probability representation, global form)**.** *For every $y \in \mathbb{R}$,*

$$F(y) = \mathbb{P}\left(\sum_{n\geq 0} \frac{U_n}{2^{n+1}} \leq y\right), \tag{57}$$

where the U_n are independent uniform variables on $[0, 1]$. Equivalently,

$$\mathrm{up}(x) = \mu((-\infty, 1 - |x|]) \tag{58}$$

for every $x \in \mathbb{R}$.

7 The survival-function Fubini identity

A particularly useful bridge, absent from the main article's probability chapter, identifies integration against the Fabius law with an ordinary integral against up. It is the compact support version of the layer-cake or tail-integration formula.

For $t \geq 0$, support and (56) give

$$\mu((t, \infty)) = 1 - F(t) = \mathrm{up}(t). \tag{59}$$

The equality remains correct for $t > 1$, when both sides are zero.

Theorem 7.1 (Expectation from the survival function)**.** *Let $g, g' : \mathbb{R} \to \mathbb{R}$ be continuous and suppose*

$$g(x) - g(0) = \int_0^x g'(t)\,\mathrm{d}t \qquad (0 \leq x \leq 1). \tag{60}$$

Then

$$\int_{[0,1]} g(x)\,\mathrm{d}\mu(x) = g(0) + \int_0^1 g'(t)\,\mathrm{up}(t)\,\mathrm{d}t. \tag{61}$$

Proof. Put

$$A = \{(t, x) \in [0, 1] \times \mathbb{R} : t < x\}, \qquad K(t, x) = \mathbf{1}_A(t, x) g'(t). \tag{62}$$

Continuity of g' on the compact interval makes K integrable with respect to $\mathrm{Leb}_{[0,1]} \otimes \mu$. Integrating first in t and using (60) gives, for $x \in [0, 1]$,

$$\int_0^1 K(t, x)\,\mathrm{d}t = \int_0^x g'(t)\,\mathrm{d}t = g(x) - g(0). \tag{63}$$

Integrating first in x gives

$$\int K(t, x)\,\mathrm{d}\mu(x) = g'(t)\mu((t, \infty)) = g'(t)\,\mathrm{up}(t) \tag{64}$$

by (59). Fubini equates the two iterated integrals. Since μ is a probability measure supported on $[0, 1]$, the integral of the constant $g(0)$ is $g(0)$, and rearrangement yields (61). ☐

Remark 7.2 (Endpoint conventions disappear under integration). The survival function uses the open ray (t, ∞). The CDF uses $(-\infty, t]$. Their complements are exact, and no endpoint atom enters because μ has a continuous CDF. Likewise, replacing $(0, 1)$ by any of the standard half-open versions changes no Lebesgue integral. The formal proof nevertheless records these null-set conversions explicitly.

8 Raw, endpoint, and tilted Laplace moments

The main article develops half moments and the negative generating function from the analytic side. The new probability–Laplace module proves that those quantities are exactly the moments of the product law.

8.1 Three moment families

For a probability measure ν supported on $[0, 1]$, define

$$U_n(\nu) = \int_{[0,1]} (1-x)^n \,\mathrm{d}\nu(x), \tag{65}$$

$$L_k(\nu; s) = \int_{[0,1]} \mathrm{e}^{-sx} x^k \,\mathrm{d}\nu(x), \tag{66}$$

$$J_k(s) = \int_0^1 t^k \,\mathrm{up}(t)\mathrm{e}^{-st} \,\mathrm{d}t. \tag{67}$$

For real s, the integrand in (66) is nonnegative, so

$$L_k(\nu; s) \geq 0. \tag{68}$$

Apply theorem 7.1 to $g(x) = \mathrm{e}^{-sx}x^k$. For $k = 0$ one obtains

$$L_0(\mu; s) = 1 - sJ_0(s). \tag{69}$$

For $k \geq 0$, applying it to $g(x) = \mathrm{e}^{-sx}x^{k+1}$ gives

$$L_{k+1}(\mu; s) = (k+1)J_k(s) - sJ_{k+1}(s). \tag{70}$$

These equations are exactly the defining clauses of `fabiusLaplaceMoment`.

Theorem 8.1 (Law moments equal analytic Fabius moments). *For every $k \in \mathbb{N}$ and $s \in \mathbb{R}$,*

$$L_k(\mu; s) = M_k(s), \tag{71}$$

where

$$M_0(s) = G(-s), \tag{72}$$

$$M_{k+1}(s) = (k+1)J_k(s) - sJ_{k+1}(s) \tag{73}$$

are the tilted Fabius moments and G is the generating function of the main article. In particular $M_k(s) \geq 0$ for real s.

Proof. The case $k = 0$ is the integration-by-parts identity (69), which is also the negative generating-function formula. The successor case is (70). □

8.2 Differential recurrence

Differentiation under the compact integral in (67) is justified by a uniform exponential majorant in a neighborhood of each real s. It gives

$$J_k'(s) = -J_{k+1}(s). \tag{74}$$

Substituting into (72)–(73) yields the cleaner recurrence

$$M_k'(s) = -M_{k+1}(s). \tag{75}$$

Iteration gives

$$M_k^{(n)}(s) = (-1)^n M_{k+n}(s). \tag{76}$$

Thus all M_k are C^∞ on $\mathbb{R}$, and the negative generating function is completely monotone:

$$(-1)^n \frac{\mathrm{d}^n}{\mathrm{d}s^n} G(-s) = M_n(s) \geq 0. \tag{77}$$

8.3 Reflection and the half moments

Reflection invariance (54) gives

$$\int (1-x)^n \,\mathrm{d}\mu(x) = \int x^n \,\mathrm{d}\mu(x). \tag{78}$$

At zero tilt the analytic moment normalization is the exact rational half-moment sequence d_n:

$$M_n(0) = d_n. \tag{79}$$

Combining the identifications yields the full probability interpretation.

Theorem 8.2 (Probability meaning of the half moments)**.** *For every $n \in \mathbb{N}$,*

$$d_n = \mathbb{E}[X^n] = \mathbb{E}[(1-X)^n] = \int_{[0,1]} x^n \,\mathrm{d}\mu(x) = \int_{[0,1]} (1-x)^n \,\mathrm{d}\mu(x). \tag{80}$$

Moreover

$$\left.\frac{\mathrm{d}^n}{\mathrm{d}s^n} G(-s)\right|_{s=0} = (-1)^n d_n. \tag{81}$$

Proof. Use (71) at $s = 0$, then (79) and (78). The derivative identity is (76) with $k = 0$ and $s = 0$. □

The equality $d_n = \mathbb{E}[X^n]$ is not merely an interpretation of a previously known rational sequence. It is the bridge that licenses positivity, log-convexity, moment-matrix positivity, and Laplace comparison arguments without rebuilding them from the recurrence for d_n. The formal library keeps the bridge in a separate module precisely so the probabilistic and saddle branches can import it independently.

---

9 From atomic laws to half-endpoint histograms

The main article introduces the polynomial step approximants $\mathcal{W}_n$ and then moves quickly to their convergence. The formal proof has a nontrivial middle layer: the coefficients first define an atomic probability measure; each atom is smeared across a centered cell; interval integrals of the resulting histogram are sandwiched between masses of slightly shrunken and enlarged intervals.

9.1 Half-endpoint indicators are ordinary indicators almost everywhere

For $a < b$, define

$$\mathbf{1}^*_{[a,b]}(x) = \begin{cases} 1, & a < x < b, \\ \frac{1}{2}, & x = a \text{ or } x = b, \\ 0, & \text{otherwise.} \end{cases} \tag{82}$$

The two endpoint values are essential for pointwise gluing of adjacent cells, but irrelevant to Lebesgue integration.

Lemma 9.1 (Almost-everywhere replacement). *For every $a, b \in \mathbb{R}$,*

$$\mathbf{1}^*_{[a,b]} = \mathbf{1}_{(a,b]} \quad \textit{Lebesgue-almost everywhere.} \tag{83}$$

Consequently

$$\int_{\mathbb{R}} \mathbf{1}^*_{[a,b]}(x)\,\mathrm{d}x = \max(b-a, 0), \tag{84}$$

and, for $c \le d$,

$$\int_c^d \mathbf{1}^*_{[a,b]}(x)\,\mathrm{d}x = \max\bigl(\min(d,b) - \max(c,a), 0\bigr). \tag{85}$$

Proof. The functions differ only at a and b, a null set. The first integral is the length of $(a, b]$. The second is the length of $(c, d] \cap (a, b] = (\max(c, a), \min(d, b)]$, with truncation at zero when the interval is empty. □

9.2 Atomic measure, cell geometry, and exact normalization

At level n, let $x_{n,m}$ be the location of the mth polynomial atom and let $p_{n,m} \ge 0$ be its normalized mass, with

$$\sum_m p_{n,m} = 1. \tag{86}$$

The corresponding atomic law is

$$\nu_n = \sum_m p_{n,m} \delta_{x_{n,m}}. \tag{87}$$

Let

$$h_n = 2^{-n-1} \tag{88}$$

be the half-width of a cell and

$$C_{n,m} = [x_{n,m} - h_n, x_{n,m} + h_n]. \tag{89}$$

Its full width is $2h_n = 2^{-n}$. Smearing the atom uniformly over its cell produces the histogram

$$\mathcal{W}_n(x) = 2^n \sum_m p_{n,m} \mathbf{1}^*_{C_{n,m}}(x). \tag{90}$$

By (84), each term has integral $p_{n,m}$, hence

$$\int_{\mathbb{R}} \mathcal{W}_n(x)\,\mathrm{d}x = 1. \tag{91}$$

The pointwise half weights ensure that when two cells share an endpoint, the two half contributions add to the correct plateau value.

9.3 The overlap sandwich

For a fixed interval $[a, b]$ with $a \le b$, let

$$\ell_{n,m}(a, b) = |[a, b] \cap C_{n,m}| = \max\bigl(\min(b, x_{n,m} + h_n) - \max(a, x_{n,m} - h_n), 0\bigr). \tag{92}$$

Then

$$\int_a^b \mathcal{W}_n(x)\,\mathrm{d}x = \sum_m p_{n,m}\, 2^n \ell_{n,m}(a, b). \tag{93}$$

The normalized overlap $2^n \ell_{n,m}$ lies in $[0, 1]$.

If $x_{n,m} \in (a + h_n, b - h_n]$, then the entire cell lies inside $[a, b]$, so the normalized overlap is 1. If the cell meets $[a, b]$, then its center lies in $(a - h_n, b + h_n]$. Therefore, term by term,

$$\mathbf{1}_{(a+h_n,b-h_n]}(x_{n,m}) \le 2^n \ell_{n,m}(a, b) \le \mathbf{1}_{(a-h_n,b+h_n]}(x_{n,m}). \tag{94}$$

Multiplication by $p_{n,m}$ and summation gives the central bridge.

Theorem 9.2 (Atomic–histogram interval sandwich)**.** *For $a \le b$,*

$$\nu_n((a + h_n, b - h_n]) \le \int_a^b \mathcal{W}_n(x)\,\mathrm{d}x \le \nu_n((a - h_n, b + h_n]). \tag{95}$$

This is the exact content of the inner- and outer-overlap lemmas in `StepMeasureBridge.lean`. Its importance is conceptual: it converts weak convergence of measures into convergence of integrals of discontinuous histograms without claiming pointwise convergence of densities from weak convergence alone.

9.4 Portmanteau plus a moving-boundary squeeze

The atomic laws ν_n coincide with the finite convolution laws and converge weakly to the absolutely continuous law with density up. Because that limit gives zero mass to every singleton, the frontier of $(a, b]$ is null. Portmanteau therefore gives

$$\nu_n((a, b]) \longrightarrow \int_a^b \mathrm{up}(x)\,\mathrm{d}x. \tag{96}$$

The intervals in (95) move with n, so one more squeeze is needed. Fix $\varepsilon > 0$ and choose $\delta > 0$ with $4\delta < \varepsilon$. Eventually $h_n \le \delta$, and hence

$$(a + \delta, b - \delta] \subset (a + h_n, b - h_n], \tag{97}$$

$$(a - h_n, b + h_n] \subset (a - \delta, b + \delta]. \tag{98}$$

The bound $0 \le \mathrm{up} \le 1$ gives

$$0 \le \int_a^b \mathrm{up} - \int_{a+\delta}^{b-\delta} \mathrm{up} \le 2\delta, \tag{99}$$

$$0 \le \int_{a-\delta}^{b+\delta} \mathrm{up} - \int_a^b \mathrm{up} \le 2\delta. \tag{100}$$

Apply (96) to the two fixed comparison intervals and combine with (95).

Theorem 9.3 (Interval-integral convergence of the step approximants)**.** *For every $a, b \in \mathbb{R}$,*

$$\int_a^b \mathcal{W}_n(x)\,\mathrm{d}x \longrightarrow \int_a^b \mathrm{up}(x)\,\mathrm{d}x. \tag{101}$$

Proof. The preceding fixed-δ sandwich proves the result for $a \le b$. If $b < a$, reverse the orientation of both interval integrals. $\square$

Warning 9.4 (The logical order cannot be shortened). Weak convergence of ν_n does not by itself imply pointwise convergence of the histograms $\mathcal{W}_n$, nor even convergence of their values at a single point. The formal route is

$$\begin{aligned}\text{weak convergence} &\implies \text{null-boundary interval masses}\\ &\implies \text{moving-cell sandwich} \implies \text{interval-integral convergence.}\end{aligned}$$

Other parts of the development then use shape, refinement, Fourier decay, or difference quotients to obtain stronger function-level convergence.

Part III
The reusable discrete engines

10 Weighted paths solve triangular recurrences

The exact inverse-power recurrence for $F(2^{-n})$ is triangular. The main article states its composition formula, but the Lean proof first establishes a generic solver for every semiring-valued triangular recurrence. That abstraction explains both the form of the answer and the uniqueness argument.

10.1 Compositions as increasing paths

A composition of n is an ordered list of positive integers

$$c = (r_1, \ldots, r_m), \qquad r_1 + \cdots + r_m = n. \tag{102}$$

Its partial sums

$$s_0 = 0, \qquad s_j = r_1 + \cdots + r_j \quad (1 \le j \le m) \tag{103}$$

form a strictly increasing path

$$0 = s_0 < s_1 < \cdots < s_m = n. \tag{104}$$

Conversely, every such path determines the positive increments $r_j = s_j - s_{j-1}$.

Let R be a semiring and let

$$w : \mathbb{N} \times \mathbb{N} \to R \tag{105}$$

be an edge-weight function. Define

$$W_w(c) = \prod_{j=1}^{m} w(s_{j-1}, s_j), \tag{106}$$

$$P_w(n) = \sum_{c \models n} W_w(c). \tag{107}$$

For $n = 0$, there is one empty composition and its empty product is 1, so

$$P_w(0) = 1. \tag{108}$$

10.2 Last-edge decomposition

Every nonempty path to n has a unique last predecessor $k < n$. Deleting its final edge gives a path to k, and appending the edge $k \to n$ reverses the operation. Therefore

Theorem 10.1 (Last-edge decomposition). *For $n > 0$,*

$$P_w(n) = \sum_{k=0}^{n-1} P_w(k)w(k, n). \tag{109}$$

Proof. Partition the finite set of compositions of n by the sum k of all blocks except the last. For fixed k, deleting the final block is a bijection with compositions of k. Path-weight multiplicativity gives

$$W_w(c \parallel (n - k)) = W_w(c)w(k, n).$$

Summing first over compositions of k and then over $k < n$ gives (109). □

One can instead partition by the number of edges. Let $P_{w,r}(n)$ be the contribution of compositions with exactly r blocks. Uniformly in n,

$$P_w(n) = \sum_{r=0}^{n} P_{w,r}(n), \tag{110}$$

and for $n > 0$ the zero-edge term vanishes, so the range may be written $1 \leq r \leq n$.

10.3 The generic triangular solver

Theorem 10.2 (Path-sum solution of a triangular recurrence). *Let $x : \mathbb{N} \to R$ satisfy*

$$x_0 = b, \tag{111}$$

$$x_n = \sum_{k=0}^{n-1} x_k w(k, n) \qquad (n > 0). \tag{112}$$

Then

$$x_n = bP_w(n) \qquad (n \geq 0). \tag{113}$$

Consequently the recurrence and initial value determine x uniquely.

Proof. Use strong induction on n. At $n = 0$, (108) gives $bP_w(0) = b = x_0$. For $n > 0$, insert the induction hypotheses into (112):

$$x_n = \sum_{k<n} bP_w(k)w(k, n) = b \sum_{k<n} P_w(k)w(k, n) = bP_w(n),$$

where the last equality is theorem 10.1. Any two solutions therefore equal the same path sum at each index. □

Remark 10.3. No subtraction, division, topology, or order is used. The theorem works in an arbitrary semiring. This is why the formal development separates it from the rational Fabius specialization.

11 The nonrecursive composition formula for $F(2^{-n})$

Let a_n be the normalized inverse-dyadic recurrence sequence used in the main article, so

$$F(2^{-n}) = 2^{-\binom{n}{2}} a_n. \tag{114}$$

Its triangular recurrence has edge weights

$$w(k,n) = \frac{1}{(2^n - 1)(n-k+1)!} \qquad (0 \le k < n). \tag{115}$$

Because $a_0 = 1$, theorem 10.2 gives $a_n = P_w(n)$.

For a composition $c = (r_1, \ldots, r_m)$ of n, the edge from s_{j-1} to s_j has $n-k$ replaced by r_j, hence

$$w(s_{j-1}, s_j) = \frac{1}{(2^{s_j} - 1)(r_j + 1)!}. \tag{116}$$

Therefore:

Theorem 11.1 (Nonrecursive inverse-dyadic formula). *For every $n \in \mathbb{N}$,*

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{(r_1,\ldots,r_m) \models n} \prod_{j=1}^{m} \frac{1}{(2^{r_1+\cdots+r_j} - 1)(r_j+1)!}. \tag{117}$$

At $n = 0$, the unique empty composition contributes the empty product 1.

Proof. Apply theorem 10.2 with (115), translate paths into compositions using (103), and multiply by the outer normalization in (114). □

The formula may be grouped by the number m of blocks:

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{m=1}^{n} \sum_{\substack{r_1+\cdots+r_m=n \\ r_j \ge 1}} \prod_{j=1}^{m} \frac{1}{(2^{r_1+\cdots+r_j} - 1)(r_j+1)!}, \qquad n > 0. \tag{118}$$

11.1 Small cases as structural checks

Example 11.2 (The value at $1/2$). For $n = 1$ the sole composition is (1), with weight

$$\frac{1}{(2^1 - 1)2!} = \frac{1}{2}.$$

The outer power is $2^0 = 1$, so $F(1/2) = 1/2$.

Example 11.3 (The value at $1/4$). The two compositions of 2 are (2) and $(1,1)$:

$$W(2) = \frac{1}{(4-1)3!} = \frac{1}{18},$$
$$W(1,1) = \frac{1}{(2-1)2!} \frac{1}{(4-1)2!} = \frac{1}{12}.$$

Their sum is $5/36$ and the outer factor is 2^{-1}, giving

$$F(1/4) = \frac{5}{72}. \tag{119}$$

Example 11.4 (The value at $1/8$)**.** The four compositions of 3 have weights

$$\frac{1}{168}, \qquad \frac{1}{84}, \qquad \frac{1}{252}, \qquad \frac{1}{168},$$

for (3), $(1,2)$, $(2,1)$, and $(1,1,1)$ respectively. Their sum is $1/36$; multiplying by $2^{-\binom{3}{2}} = 1/8$ gives

$$F(1/8) = \frac{1}{288}. \tag{120}$$

The composition formula is not a mysterious closed form guessed from the first values. It is the universal unrolling of a triangular recurrence. Every block records a jump, every partial sum records the endpoint of that jump, and the product records the chronological edge weights. The formal path layer makes this explanation exact and reusable.

---

12 A finite-row Toeplitz convergence theorem

The complex-shift and discrete-limit formulas involve a row of coefficients depending on n, not a fixed summability kernel. The main article verifies the Fabius specialization but omits the abstract theorem. The needed result is a finite-row version of the Toeplitz summability principle in a normed ring.

12.1 Abstract statement

Let K be a normed ring. For each n, let J_n be a finite index set, let $w_{n,j} \in K$, and let $q(n,j) \in \mathbb{N}$ be the sampled sequence index. Assume:

$$\sum_{j \in J_n} w_{n,j} = 1, \tag{121}$$

$$\sum_{j \in J_n} \|w_{n,j}\| \le C \tag{122}$$

for one constant $C \ge 0$, and assume that the sampled indices move uniformly into every tail:

$$\forall N\ \exists n_0\ \forall n \ge n_0\ \forall j \in J_n, \qquad q(n,j) \ge N. \tag{123}$$

Theorem 12.1 (Finite-row Toeplitz convergence)**.** *If $H_m \to L$ in K, then*

$$\sum_{j \in J_n} w_{n,j} H_{q(n,j)} \longrightarrow L. \tag{124}$$

Proof. Fix $\varepsilon > 0$. Since $H_m \to L$, choose N such that

$$\|H_m - L\| < \frac{\varepsilon}{C+1} \qquad (m \ge N). \tag{125}$$

Choose n_0 from (123). For $n \ge n_0$, use the row-mass identity to subtract L:

$$\sum_{j \in J_n} w_{n,j} H_{q(n,j)} - L = \sum_{j \in J_n} w_{n,j} \big(H_{q(n,j)} - L\big).$$

Therefore

$$\begin{aligned}\left\| \sum_{j\in J_n} w_{n,j} H_{q(n,j)} - L \right\| &\le \sum_{j\in J_n} \|w_{n,j}\| \, \|H_{q(n,j)} - L\| \\ &< \frac{\varepsilon}{C+1} \sum_{j\in J_n} \|w_{n,j}\| \\ &\le \varepsilon \frac{C}{C+1} < \varepsilon.\end{aligned}$$

This proves convergence. □

Remark 12.2. The weights need not be positive, the rows need not be nested, and no entrywise limit of $w_{n,j}$ is required. The theorem uses exactly three facts: row mass one, uniformly bounded total variation, and uniform migration of all sampled indices into the tail.

13 The Fabius Toeplitz rows and corrected discrete prefixes

13.1 The half-base q-binomial weights

For $0 \le j \le n$, define

$$w_{n,j} = \frac{(-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} (1/2)^{\binom{j+1}{2}}}{(1/2;1/2)_n}. \tag{126}$$

The half-base finite q-binomial theorem evaluated at $z = 1/2$ gives

$$\sum_{j=0}^{n} w_{n,j} = 1. \tag{127}$$

Taking absolute values removes the signs and changes the finite product from $(1/2;1/2)_n$ to $(-1/2;1/2)_n$:

$$\sum_{j=0}^{n} |w_{n,j}| = \frac{(-1/2;1/2)_n}{(1/2;1/2)_n}. \tag{128}$$

The formal proof uses the elementary uniform lower bound

$$(1/2;1/2)_n \ge \frac{1}{4} \tag{129}$$

and the product comparison

$$(-1/2;1/2)_n (1/2;1/2)_n \le 1. \tag{130}$$

Hence

$$\sum_{j=0}^{n} |w_{n,j}| \le 16. \tag{131}$$

The constant 16 is deliberately coarse but uniform and algebraically simple.

The reindexing used by the discrete formula is

$$q(n,j) = 2n - j. \tag{132}$$

Since $0 \le j \le n$,

$$q(n,j) = 2n - j \ge n. \tag{133}$$

Thus every index in row n lies beyond the common threshold n.

Corollary 13.1 (Specialized Fabius row convergence). *For every convergent sequence $H_m \to L$ in $\mathbb{R}$ or $\mathbb{C}$,*

$$\sum_{j=0}^{n} w_{n,j} H_{2n-j} \longrightarrow L. \tag{134}$$

Proof. Apply theorem 12.1 with $C = 16$ and use (127), (131), and (133). □

13.2 Why the inner range uses a half-cell correction

The finite discrete approximant contains an inner prefix whose source-facing notation has inclusive upper endpoint

$$\left\lfloor 2^p x - \frac{1}{2} \right\rfloor. \tag{135}$$

A direct natural-number range cannot represent the value -1, which should mean an empty sum. The safe range length is

$$\ell_p(x) = \left\lfloor 2^p x + \frac{1}{2} \right\rfloor_{\mathbb{N}}, \tag{136}$$

where the natural floor clamps negative results to zero. For $x \geq 0$,

$$\ell_p(x) = \left\lfloor 2^p x - \frac{1}{2} \right\rfloor + 1, \tag{137}$$

so the natural range $0 \leq r < \ell_p(x)$ exactly represents the intended inclusive sum. Moreover

$$\ell_p(x) = 0 \iff 2^p x < \frac{1}{2}. \tag{138}$$

In particular every prefix is empty for $x \leq 0$, so every finite approximant vanishes there before taking a limit.

Warning 13.2 (The half-cell is not cosmetic). Replacing (136) by $\lfloor 2^p x \rfloor$ shifts every jump and breaks the centered nearest-integer convention. Replacing it by a signed inclusive endpoint inside a natural range mishandles the empty case. The formula $\lfloor 2^p x + 1/2 \rfloor_{\mathbb{N}}$ simultaneously fixes both issues and is the exact convention used by the formal development.

13.3 Where the abstract theorem enters the discrete limit

After the algebraic q-binomial reindexing, the nth discrete approximant takes the form

$$A_n(x) = \sum_{j=0}^{n} w_{n,j} H_{2n-j}(x), \tag{139}$$

where, for each fixed x, the inner objects $H_p(x)$ converge to the desired translated Fabius value. The entire outer-limit argument is then

$$H_p(x) \to L(x) \quad + \quad (127), (131), (133) \quad \Longrightarrow \quad A_n(x) \to L(x).$$

All Fabius-specific work lies in identifying H_p and its limit; all row summation is contained in theorem 12.1.

Part IV
The formal and analytic machinery of the all-orders saddle expansion

14 Parameter-dependent Poincaré expansions

The endpoint asymptotic is not an ordinary expansion with constant coefficients. Its coefficients retain a bounded one-periodic dependence on the exact Lambert phase. The formal library therefore uses an asymptotic API in which coefficient families may depend on the asymptotic parameter, provided that dependence remains bounded.

Let ℓ be a filter on a parameter space A, let $\sigma : A \to \mathbb{K}$ be a scale tending to zero, and let $f : A \to E$ take values in a normed vector space over the normed field $\mathbb{K}$. For coefficient families $c_k : A \to E$, put

$$S_N[f](a) = \sum_{k=0}^{N-1} \sigma(a)^k c_k(a). \tag{140}$$

Definition 14.1 (Full parameter-dependent expansion)**.** We write

$$f(a) \sim_\ell \sum_{k\geq 0} \sigma(a)^k c_k(a) \tag{141}$$

when

1. every c_k is bounded along ℓ, equivalently $c_k = O_\ell(1)$;
2. for every $N \in \mathbb{N}$,
$$f - S_N[f] = O_\ell(\sigma^N). \tag{142}$$

This is `HasAsymptoticExpansion`. Boundedness of the coefficients is not redundant: if arbitrary unbounded parameter dependence were allowed, powers of σ could be moved between the "coefficient" and the scale, destroying the meaning of order.

14.1 Elementary algebra of full expansions

Suppose f and g have expansions with the same scale. Direct manipulation of finite partial sums proves:

$$f + g \sim \sum_{k\geq 0} \sigma^k (c_k + d_k), \tag{143}$$

$$f - g \sim \sum_{k\geq 0} \sigma^k (c_k - d_k), \tag{144}$$

$$\alpha f \sim \sum_{k\geq 0} \sigma^k (\alpha c_k). \tag{145}$$

The API also supports transport across eventual equality and pullback along a map tending to the source filter. These apparently routine lemmas are what let the final proof switch between complex and real saddle ratios, replace exact identities after a threshold, and move from the logarithmic coordinate to $x \downarrow 0$ without reopening every remainder estimate.

14.2 Flat functions

Definition 14.2 (Flat relative to a scale)**.** A function r is flat relative to σ along ℓ if

$$r = O_\ell(\sigma^N) \qquad \text{for every } N \in \mathbb{N}. \tag{146}$$

Theorem 14.3 (Flat remainders are invisible to all coefficients)**.** *A flat r has the full expansion with every coefficient zero. Consequently, if*

$$f \sim \sum_{k\geq 0} \sigma^k c_k, \tag{147}$$

then

$$f + r \sim \sum_{k\geq 0} \sigma^k c_k, \qquad f - r \sim \sum_{k\geq 0} \sigma^k c_k. \tag{148}$$

The implications hold in both directions.

Proof. For the zero-coefficient expansion, the Nth partial sum is identically zero, so the remainder condition is exactly (146). Add or subtract this expansion coefficientwise from the expansion of f. Reversibility follows by moving r to the other side. □

This theorem is the formal reason that the forward negative-Laplace tail can be restored after the central saddle expansion without changing any algebraic-order coefficient.

14.3 Exact logarithmic coordinates

Define

$$x(t) = 2^{-t}, \qquad t(x) = -\log_2 x \quad (x > 0). \tag{149}$$

These maps are exact inverses on $x > 0$, and

$$t \to +\infty \iff x(t) \downarrow 0, \qquad x \downarrow 0 \iff t(x) \to +\infty. \tag{150}$$

Theorem 14.4 (Full-expansion coordinate equivalence)**.** *For functions f, σ, c_k on the positive half-line,*

$$f(2^{-t}) \sim_{t\to\infty} \sum_{k\geq 0} \sigma(2^{-t})^k c_k(2^{-t}) \tag{151}$$

if and only if

$$f(x) \sim_{x\downarrow 0} \sum_{k\geq 0} \sigma(x)^k c_k(x). \tag{152}$$

Proof. Pull back the full expansion along either coordinate map. The two composites are eventually identical on their respective filters by (149); boundedness and every remainder estimate therefore transfer in both directions. □

15 Recursive coefficients of a formal exponential

The saddle integrand contains the exponential of a formal perturbation. Rather than invoke Bell polynomials or repeatedly expand a universal power series, the Lean development uses a finite recurrence whose nth output depends only on the first n input coefficients.

Let R be a commutative $\mathbb{Q}$-algebra, and let

$$E(u) = \sum_{j\geq 1} E_j u^j \tag{153}$$

be a formal exponent with zero constant term. Define $a_n \in R$ recursively by

$$a_0 = 1, \tag{154}$$

$$a_{n+1} = \frac{1}{n+1}\sum_{j=0}^{n}(j+1)E_{j+1}a_{n-j}. \tag{155}$$

Theorem 15.1 (Correctness and uniqueness of the exponential recurrence). *The power series*

$$A(u) = \sum_{n\geq 0} a_n u^n \tag{156}$$

satisfies

$$A'(u) = E'(u)A(u), \qquad A(0) = 1. \tag{157}$$

It is the unique formal power series with these two properties, and therefore

$$A(u) = \exp(E(u)). \tag{158}$$

Proof. The coefficient of u^n in A' is $(n+1)a_{n+1}$. The coefficient of u^n in $E'A$ is

$$\sum_{j=0}^{n}(j+1)E_{j+1}a_{n-j},$$

so (157) is coefficientwise equivalent to (155). Conversely the differential equation and initial value determine a_0, then a_1, then a_2, and so on, proving uniqueness. The universal formal exponential satisfies the same equation and initial value. □

The first coefficients are

$$a_1 = E_1, \tag{159}$$

$$a_2 = E_2 + \frac{1}{2}E_1^2, \tag{160}$$

$$a_3 = E_3 + E_1E_2 + \frac{1}{6}E_1^3, \tag{161}$$

$$a_4 = E_4 + E_1E_3 + \frac{1}{2}E_2^2 + \frac{1}{2}E_1^2E_2 + \frac{1}{24}E_1^4. \tag{162}$$

Equation (162) is the exact finite identity used for the second saddle correction.

15.1 Structural laws

The recurrence is preferable to a black-box exponential because it exposes several laws needed downstream.

Proposition 15.2 (Finite dependence). *The coefficient a_n depends only on $E_1, \ldots, E_n$. If two exponent families agree through order n, their nth exponential coefficients agree.*

Proposition 15.3 (Naturality). *For every morphism of commutative $\mathbb{Q}$-algebras $\varphi : R \to S$,*

$$\varphi(a_n(E)) = a_n(\varphi \circ E). \tag{163}$$

In particular, evaluation of polynomial- or function-valued coefficients commutes with the recurrence.

Proposition 15.4 (Rescaling, sums, and parity). *Let $c \in R$ be central.*

1. *If* $\widetilde{E}_j = c^j E_j$, *then*
$$a_n(\widetilde{E}) = c^n a_n(E). \tag{164}$$

2. *If* E *and* F *are two exponent families, then*
$$a_n(E+F) = \sum_{r+s=n} a_r(E)a_s(F). \tag{165}$$

3. *If* $E_j(-v) = (-1)^j E_j(v)$, *then*
$$a_n(-v) = (-1)^n a_n(v). \tag{166}$$

Proof architecture. Naturality is induction through finite sums and rational scalar multiplication. Rescaling follows from the recurrence because every summand has total degree $(j+1)+(n-j) = n+1$. The addition law is the coefficient form of $\exp(E+F) = \exp(E)\exp(F)$. Parity is the specialization $c = -1$ of rescaling after pointwise reflection in v. □

16 Recursive coefficients of a formal logarithm

Let
$$A(u) = 1 + \sum_{n\geq 1} a_n u^n. \tag{167}$$
We seek
$$B(u) = \log A(u) = \sum_{n\geq 1} b_n u^n. \tag{168}$$
Instead of expanding the universal logarithm, use
$$A(u)B'(u) = A'(u). \tag{169}$$
Solving the coefficient of u^n for b_{n+1} gives
$$b_0 = 0, \tag{170}$$
$$b_{n+1} = a_{n+1} - \frac{1}{n+1}\sum_{j=0}^{n-1}(n-j)b_{n-j}a_{j+1}. \tag{171}$$
The empty sum at $n = 0$ gives $b_1 = a_1$.

Theorem 16.1 (Correctness and uniqueness of the logarithm recurrence). *If* $a_0 = 1$, *the series generated by* (171) *is the unique series with zero constant term satisfying* (169)*; hence it is the formal logarithm of* A.

Proof. The coefficient of u^n in AB' contains the term $(n+1)b_{n+1}$ from $a_0 b'$, plus terms involving only $b_1, \dots, b_n$. Equating it with $(n+1)a_{n+1}$ and dividing by $n+1$ gives (171). This triangular relation proves uniqueness. The formal logarithm satisfies $B(0) = 0$ and $B'A = A'$. □

The first coefficients are
$$b_1 = a_1, \tag{172}$$
$$b_2 = a_2 - \frac{1}{2}a_1^2, \tag{173}$$
$$b_3 = a_3 - a_1a_2 + \frac{1}{3}a_1^3, \tag{174}$$
$$b_4 = a_4 - a_1a_3 - \frac{1}{2}a_2^2 + a_1^2a_2 - \frac{1}{4}a_1^4. \tag{175}$$

As with the exponential recurrence, b_n depends only on $a_1, \dots, a_n$, commutes with $\mathbb{Q}$-algebra morphisms and evaluation, obeys the rescaling law

$$a_j \mapsto c^j a_j \quad \Longrightarrow \quad b_n \mapsto c^n b_n, \tag{176}$$

and preserves parity.

Remark 16.2 (Why a_0 is a hypothesis, not an input). The recursive definition (171) never reads a_0; it is meaningful for every sequence. Correctness as a logarithm, however, requires $a_0 = 1$. Separating the recurrence from its normalization makes finite coefficient calculations easier while keeping the theorem honest.

16.1 From mass expansions to logarithmic expansions

Suppose a positive real function has a full expansion

$$R(a) \sim 1 + \sum_{n \geq 1} \sigma(a)^n a_n(a), \qquad \sigma(a) \to 0, \tag{177}$$

with bounded coefficient families. Finite polynomial remainders, discussed in the next section, show that for each N,

$$\log R(a) = \sum_{n=1}^{N-1} \sigma(a)^n b_n(a) + O(\sigma(a)^N), \tag{178}$$

where b_n is generated by (171). The positivity hypothesis fixes the real branch and is eventually automatic when $R \to 1$.

17 Exact finite remainders from formal coefficient identities

Formal power-series equality says that two expressions have the same coefficients below a chosen order. An asymptotic proof needs an evaluated identity with an explicit factor of the first omitted power. The module `SaddleExpansionFiniteRemainder.lean` provides exactly that bridge without invoking an infinite series.

Fix $L \geq 1$ and truncate the exponent:

$$E_{<L}(X) = \sum_{k=1}^{L-1} E_k X^k. \tag{179}$$

Truncate the scalar exponential polynomial at the same order and substitute:

$$T_L(X) = \sum_{q=0}^{L-1} \frac{E_{<L}(X)^q}{q!}. \tag{180}$$

Let

$$A_{<L}(X) = \sum_{k=0}^{L-1} a_k X^k \tag{181}$$

be the polynomial formed from the recursive exponential coefficients.

Theorem 17.1 (Exact divisibility of the finite defect). *If $E_0 = 0$, then*

$$X^L \mid \big(T_L(X) - A_{<L}(X)\big). \tag{182}$$

Equivalently, there is a canonical polynomial $Q_L(X)$ such that

$$T_L(X) = A_{<L}(X) + X^L Q_L(X). \tag{183}$$

Proof. The coefficient of X^k in T_L agrees with that of the full formal substitution $\exp(E(X))$ for every $k < L$: terms of scalar exponential degree $q \geq L$ cannot contribute below degree L because E has zero constant term, and truncating E itself does not alter coefficients below L. By theorem 15.1, that common coefficient is a_k. Hence the first L coefficients of the defect vanish, which is equivalent to divisibility by X^L. The formal construction takes Q_L to be the result of applying polynomial division by X exactly L times. ☐

Evaluating at $x \in R$ gives the identity actually used in estimates:

$$\sum_{q=0}^{L-1} \frac{1}{q!} \left(\sum_{k=1}^{L-1} E_k x^k \right)^q = \sum_{k=0}^{L-1} a_k x^k + x^L Q_L(x). \tag{184}$$

The quotient Q_L is a polynomial in x whose coefficients are algebraic expressions in the finitely many parameters $E_1, \dots, E_{L-1}$. If those parameter families are uniformly bounded, then Q_L is uniformly bounded on a fixed compact x-window. Thus the exact factorization (184) turns formal coefficient agreement into a uniform $O(x^L)$ estimate with no appeal to convergence of an infinite formal series.

18 Gaussian contraction: integration becomes polynomial algebra

After the saddle variable is scaled by $v = \sqrt{\lambda}\,\theta$, the leading kernel is the standard Gaussian

$$\gamma(v) = \mathrm{e}^{-v^2/2}. \tag{185}$$

Every finite reference term is $\gamma(v)$ times a complex polynomial. The generic contraction module turns its whole-line integral into an algebraic linear functional.

18.1 Gaussian moments

Define the unnormalized moments

$$M_n^{\mathrm{G}} = \int_{\mathbb{R}} \mathrm{e}^{-v^2/2} v^n \,\mathrm{d}v. \tag{186}$$

Every moment is absolutely integrable. Symmetry gives

$$M_{2j+1}^{\mathrm{G}} = 0. \tag{187}$$

Integration by parts, with $u = v^{n+1}$ and $\mathrm{d}w = -v\mathrm{e}^{-v^2/2}\,\mathrm{d}v$, gives

$$M_{n+2}^{\mathrm{G}} = (n+1) M_n^{\mathrm{G}}. \tag{188}$$

Since

$$M_0^{\mathrm{G}} = \sqrt{2\pi}, \tag{189}$$

the normalized moments

$$\mu_n = \frac{M_n^{\mathrm{G}}}{\sqrt{2\pi}} \tag{190}$$

satisfy

$$\mu_0 = 1, \quad \mu_1 = 0, \quad \mu_{n+2} = (n+1)\mu_n. \tag{191}$$

Thus

$$\mu_{2j} = (2j-1)!!, \qquad \mu_{2j+1} = 0. \tag{192}$$

The first even values needed below are

$$\mu_4 = 3, \quad \mu_6 = 15, \quad \mu_8 = 105, \quad \mu_{10} = 945, \quad \mu_{12} = 10395. \tag{193}$$

18.2 The contraction functional

For $p(X) = \sum_k p_k X^k \in \mathbb{C}[X]$, define

$$\mathcal{G}[p] = \sum_k p_k \mu_k. \tag{194}$$

This is a $\mathbb{C}$-linear map $\mathbb{C}[X] \to \mathbb{C}$.

Theorem 18.1 (Gaussian polynomial contraction). *For every complex polynomial p,*

$$\int_{\mathbb{R}} \mathrm{e}^{-v^2/2} p(v)\,\mathrm{d}v = \sqrt{2\pi}\,\mathcal{G}[p]. \tag{195}$$

In particular

$$\mathcal{G}[X^{2j}] = (2j-1)!!, \qquad \mathcal{G}[X^{2j+1}] = 0. \tag{196}$$

Proof. Expand p as a finite sum of monomials, interchange that finite sum with the integral, and use (190). Integrability follows coefficientwise from the absolute Gaussian moments. □

The same coefficientwise expansion yields the norm estimate

$$|\mathcal{G}[p]| \le \sum_k |p_k| \mu_{2\lceil k/2 \rceil}, \tag{197}$$

with odd coefficients contributing zero in the exact functional. In the formal proof a slightly more uniform support sum is used so it composes smoothly with polynomial families.

Remark 18.2 (Parity is an algebraic annihilator). The vanishing of odd Gaussian moments is not merely a simplification after integration. It is the mechanism that removes every odd power of the small parameter $\varepsilon = \lambda^{-1/2}$ from the mass expansion. The exponent coefficient of order m has parity m in v, the formal exponential recurrence preserves total parity, and (196) kills every odd result. The final expansion is therefore in integer powers of λ^{-1}.

19 Uniform Gaussian tails for polynomial references

Whole-line contraction is useful only after the central-window integral has been replaced by a whole-line integral with a negligible error. The omitted Gaussian-tail modules provide a single coefficient weight that controls every polynomial reference.

19.1 A factorial coefficient weight

For $p(X) = \sum_k p_k X^k$, define

$$\mathfrak{W}(p) = \sum_{k \in \operatorname{supp} p} |p_k|\, 8k!. \tag{198}$$

It is finite and nonnegative.

Lemma 19.1 (Monomial Gaussian tail). *For $A \ge 4$ and $k \in \mathbb{N}$,*

$$\int_{|v|>A} \mathrm{e}^{-v^2/2} |v|^k\,\mathrm{d}v \le 8k!\,\frac{\mathrm{e}^{-A^2/4}}{A}. \tag{199}$$

Proof architecture. Split the kernel as $\mathrm{e}^{-v^2/2} = \mathrm{e}^{-v^2/4}\mathrm{e}^{-v^2/4}$. The factor $|v|^k \mathrm{e}^{-v^2/4}$ is controlled uniformly by a factorial bound, while the remaining Gaussian tail is estimated by integrating the derivative of $\mathrm{e}^{-v^2/4}$ and using $|v| \ge A$. The constants are chosen so the result holds simultaneously for every $k \ge 0$ once $A \ge 4$; no optimization is needed in the saddle application. □

Theorem 19.2 (Polynomial Gaussian tail)**.** *For every $p \in \mathbb{C}[X]$ and $A \geq 4$,*

$$\int_{|v|>A} \left| e^{-v^2/2} p(v) \right| \, dv \leq \mathfrak{W}(p) \frac{e^{-A^2/4}}{A}. \tag{200}$$

Proof. Use $|p(v)| \leq \sum_{k \in \operatorname{supp} p} |p_k||v|^k$, integrate the finite sum, and apply theorem 19.1 coefficient-wise. ☐

19.2 The all-orders central radius

For expansion order N and scale parameter $b \geq 1$, set

$$A_N(b) = \sqrt{32(N+1)\log b}. \tag{201}$$

Then

$$e^{-A_N(b)^2/4} = b^{-8(N+1)}. \tag{202}$$

The exponent $8(N+1)$ is much stronger than the requested algebraic order $N+1$, leaving ample room for the factor $1/A_N(b)$ and for bounded coefficient weights.

Theorem 19.3 (All-orders polynomial tail)**.** *Let $b(a) \to +\infty$ along a filter and let $p_a \in \mathbb{C}[X]$ satisfy*

$$\mathfrak{W}(p_a) = O(1). \tag{203}$$

Then for every fixed N,

$$\int_{|v|>A_N(b(a))} \left| e^{-v^2/2} p_a(v) \right| \, dv = O\big(b(a)^{-(N+1)}\big). \tag{204}$$

Proof. Eventually $A_N(b(a)) \geq 4$ and $b(a) \geq 1$. Apply theorem 19.2, substitute (202), and use

$$b^{-8(N+1)} A_N(b)^{-1} \leq b^{-(N+1)}.$$

Multiplication by the bounded family (203) preserves the big-O rate. ☐

Remark 19.4. The theorem is formulated for a moving polynomial family. This is essential: the saddle reference polynomial depends periodically on the exact phase. Periodicity and smoothness make its finitely many coefficient functions bounded, hence its factorial weight is bounded.

20 Closed saddle jets and shifted Stirling numbers

The main article states several low-order jets. The new closed-form modules identify the entire jet sequence and explain the combinatorial coefficients.

Let $L = \log 2$ and let Ψ be the smooth one-periodic fluctuation in the exact negative Laplace exponent. The bounded exponent jets $Z_n(t)$ satisfy

$$Z_0(t) = -\frac{1}{2} + \frac{\Psi'(t)}{L}, \tag{205}$$

$$Z_{n+1}(t) = \frac{Z_n'(t)}{L} - (n+1)Z_n(t) + \frac{(-1)^{n+1} n!}{L}. \tag{206}$$

Define integers $c(n,m)$ by

$$\prod_{k=1}^{n} (z-k) = \sum_{m=0}^{n} c(n,m) z^m. \tag{207}$$

With $H_n = 1 + 1/2 + \cdots + 1/n$ and $H_0 = 0$, the recurrence solves explicitly.

Theorem 20.1 (Closed form for every bounded exponent jet). *For every $n \geq 0$,*

$$Z_n(t) = (-1)^n n! \left(\frac{H_n}{L} - \frac{1}{2} \right) + \sum_{m=0}^{n} c(n,m) \frac{\Psi^{(m+1)}(t)}{L^{m+1}}. \tag{208}$$

Moreover

$$c(n,m) = s(n+1, m+1), \tag{209}$$

where $s(\cdot,\cdot)$ denotes the signed Stirling number of the first kind.

Proof. Let $D = L^{-1}\,\mathrm{d}/\,\mathrm{d}t$. The derivative-dependent part of (206) is generated by the operator product

$$(D-n)(D-(n-1))\cdots(D-1)\frac{\Psi'}{L}. \tag{210}$$

Expanding the product gives the sum in (208). The constant part C_n satisfies

$$C_{n+1} = -(n+1)C_n + \frac{(-1)^{n+1} n!}{L}, \quad C_0 = -\frac{1}{2}, \tag{211}$$

whose solution is $C_n = (-1)^n n!(H_n/L - 1/2)$, using $H_{n+1} = H_n + 1/(n+1)$. Finally, the coefficients of $\prod_{k=1}^{n}(z-k)$ satisfy

$$c(n+1,m) = c(n,m-1) - (n+1)c(n,m), \tag{212}$$

which is exactly the shifted signed-Stirling recurrence; the initial rows agree. □

The first four jets are

$$Z_0 = -\frac{1}{2} + \frac{\Psi'}{L}, \tag{213}$$

$$Z_1 = \frac{1}{2} - \frac{1}{L} - \frac{\Psi'}{L} + \frac{\Psi''}{L^2}, \tag{214}$$

$$Z_2 = -1 + \frac{3}{L} + \frac{2\Psi'}{L} - \frac{3\Psi''}{L^2} + \frac{\Psi^{(3)}}{L^3}, \tag{215}$$

$$Z_3 = 3 - \frac{11}{L} - \frac{6\Psi'}{L} + \frac{11\Psi''}{L^2} - \frac{6\Psi^{(3)}}{L^3} + \frac{\Psi^{(4)}}{L^4}. \tag{216}$$

All derivatives are evaluated at the common argument t. These formulas make boundedness, one-periodicity, and C^∞ regularity immediate.

20.1 The exponent coefficient at arbitrary order

Let $\varepsilon = \lambda^{-1/2}$ and v be the Gaussian variable. The coefficient of ε^m in the centered saddle exponent is, for $m \geq 1$,

$$E_m(t,v) = \mathrm{i}^m \left(\frac{Z_{m-1}(t)}{m!} v^m + \frac{(-1)^{m+1}}{m+2} v^{m+2} \right). \tag{217}$$

The second monomial is universal: it is independent of Ψ and of every jet. It arises when the jet slope $(-1)^m(m+1)!$, the factorial $(m+2)!$, and the extra factor $\mathrm{i}^{m+2} = -\mathrm{i}^m$ cancel.

By (208), each E_m is therefore completely explicit in terms of L^{-1} and $\Psi', \ldots, \Psi^{(m)}$. It has the parity law

$$E_m(t,-v) = (-1)^m E_m(t,v). \tag{218}$$

21 From exponent coefficients to periodic mass and log coefficients

Let $a_n(t, v)$ be the recursive exponential coefficient

$$\exp\left(\sum_{m\geq 1} \varepsilon^m E_m(t, v)\right) \sim \sum_{n\geq 0} \varepsilon^n a_n(t, v). \tag{219}$$

Parity preservation gives

$$a_n(t, -v) = (-1)^n a_n(t, v). \tag{220}$$

Define the mass coefficients by Gaussian contraction:

$$B_j(t) = \mathcal{G}[a_{2j}(t, \cdot)]. \tag{221}$$

Odd orders disappear. The normalized saddle mass therefore has the formal pattern

$$1 + \frac{B_1(t)}{\lambda} + \frac{B_2(t)}{\lambda^2} + \cdots . \tag{222}$$

Apply the logarithm recurrence to $1 + B_1 u + B_2 u^2 + \cdots$ and define

$$A_j(t) = b_j(B_0, B_1, \ldots), \qquad B_0 = 1. \tag{223}$$

Then

$$A_0 = 0, \qquad A_1 = B_1, \qquad A_2 = B_2 - \frac{1}{2} B_1^2. \tag{224}$$

Every A_j and B_j is bounded, one-periodic, and C^∞.

21.1 The highest derivative has a universal coefficient

The jet Z_{2j-1} first appears at mass order j. The formalized leading-jet theorem says that its coefficient in B_j is

$$\frac{(-1)^j}{2^j j!}. \tag{225}$$

The reason is rigid: among all monomials produced by the exponential recurrence, exactly one can contain the newest jet, namely the linear occurrence of the exponent coefficient E_{2j}. Its jet monomial is

$$\mathrm{i}^{2j} \frac{Z_{2j-1}}{(2j)!} v^{2j} = (-1)^j \frac{Z_{2j-1}}{(2j)!} v^{2j}. \tag{226}$$

Gaussian contraction multiplies by $(2j-1)!! = (2j)!/(2^j j!)$, giving (225).

Because lower B_k involve only $Z_0, \ldots, Z_{2k-1}$, every nonlinear term in the logarithm recurrence for A_j uses jets strictly below Z_{2j-1}. Thus the newest jet survives with the same coefficient. Combining with the leading term $\Psi^{(2j)}/L^{2j}$ in (208) yields the ordinary-analysis corollary

$$[\Psi^{(2j)}] A_j = \frac{(-1)^j}{2^j j! L^{2j}}. \tag{227}$$

The mass version is a named Lean theorem; the final triangular transfer to the logarithmic coefficient is presently a short prose corollary rather than a separate declaration.

22 The second periodic correction, line by line

The main article records the first correction and announces the general coefficient engine. The current development now computes the second correction completely. This is the first order at which every layer of the machinery is visible at once: four exponent polynomials, the order-four exponential coefficient, five Gaussian moments, and the mass-to-logarithm conversion.

Fix the phase t and abbreviate

$$z_j = Z_j(t). \tag{228}$$

From (217), the first four exponent polynomials are

$$E_1 = \mathrm{i}\left(z_0 X + \frac{X^3}{3}\right), \tag{229}$$

$$E_2 = -\frac{z_1}{2}X^2 + \frac{1}{4}X^4, \tag{230}$$

$$E_3 = \mathrm{i}\left(-\frac{z_2}{6}X^3 - \frac{1}{5}X^5\right), \tag{231}$$

$$E_4 = \frac{z_3}{24}X^4 - \frac{1}{6}X^6. \tag{232}$$

At order four in ε, (162) gives

$$a_4 = E_4 + E_3E_1 + \frac{1}{2}E_2^2 + \frac{1}{2}E_2E_1^2 + \frac{1}{24}E_1^4. \tag{233}$$

Every occurrence of i cancels because the odd exponent polynomials carry one common square root of -1. Expanding and collecting even monomials gives

$$a_4 = c_4X^4 + c_6X^6 + c_8X^8 + c_{10}X^{10} + c_{12}X^{12}, \tag{234}$$

where

$$c_4 = \frac{z_0^4}{24} + \frac{z_0^2 z_1}{4} + \frac{z_0 z_2}{6} + \frac{z_1^2}{8} + \frac{z_3}{24}, \tag{235}$$

$$c_6 = \frac{z_0^3}{18} - \frac{z_0^2}{8} + \frac{z_0 z_1}{6} + \frac{z_0}{5} - \frac{z_1}{8} + \frac{z_2}{18} - \frac{1}{6}, \tag{236}$$

$$c_8 = \frac{z_0^2}{36} - \frac{z_0}{12} + \frac{z_1}{36} + \frac{47}{480}, \tag{237}$$

$$c_{10} = \frac{z_0}{162} - \frac{1}{72}, \tag{238}$$

$$c_{12} = \frac{1}{1944}. \tag{239}$$

The five rational constants in these formulas are the visible trace of the universal jet-free monomials in (217).

22.1 Gaussian contraction

Using (193), the second mass coefficient is

$$B_2 = 3c_4 + 15c_6 + 105c_8 + 945c_{10} + 10395c_{12}. \tag{240}$$

A direct simplification gives

$$\begin{aligned} B_2 = {} & \frac{z_0^4}{8} + \frac{5z_0^3}{6} + \frac{3z_0^2 z_1}{4} + \frac{25z_0^2}{24} + \frac{5z_0 z_1}{2} + \frac{z_0 z_2}{2} + \frac{z_0}{12} \\ & + \frac{3z_1^2}{8} + \frac{25z_1}{24} + \frac{5z_2}{6} + \frac{z_3}{8} + \frac{1}{288}. \end{aligned} \tag{241}$$

The first mass coefficient, already logarithmic at first order, is

$$B_1 = A_1 = -\frac{z_0^2}{2} - z_0 - \frac{z_1}{2} - \frac{1}{12}. \tag{242}$$

Substitution of (213)–(214) reduces this to the compact formula from the main article,

$$A_1(t) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(t) + \Psi'(t)^2}{2L^2}. \tag{243}$$

22.2 Taking the logarithm

By (224),

$$A_2 = B_2 - \frac{1}{2}B_1^2. \tag{244}$$

The quartic and many lower-degree terms cancel. The result is

Theorem 22.1 (Closed second periodic correction)**.** *With $z_j = Z_j(t)$,*

$$\begin{aligned} A_2(t) = \frac{1}{24}\big(&8z_0^3 + 12z_0^2 z_1 + 12z_0^2 + 48z_0 z_1 + 12z_0 z_2 \\ &+ 6z_1^2 + 24z_1 + 20z_2 + 3z_3\big). \end{aligned} \tag{245}$$

It is a bounded smooth one-periodic function.

Proof. Insert (241) and (242) into (244), expand the square, and collect. Periodicity and smoothness follow from the same properties of $Z_0, \dots, Z_3$; boundedness follows from periodic continuity. □

Substitution of (213)–(216) makes A_2 an explicit differential polynomial in

$$\Psi',\ \Psi'',\ \Psi^{(3)},\ \Psi^{(4)} \tag{246}$$

with coefficients in $\mathbb{Q}[L^{-1}]$. The jet form (245) is shorter, more stable under formal manipulation, and displays the triangular structure that generalizes to all orders.

At order two the mass coefficient (241) contains a quartic term $z_0^4/8$. It disappears from A_2 because the logarithm removes the disconnected square $B_1^2/2$. This is the same connected-versus-disconnected cancellation familiar from cumulants. The formal logarithm recurrence performs that cancellation over an arbitrary commutative $\mathbb{Q}$-algebra, before any analytic interpretation is imposed.

23 Assembly of the full endpoint expansion

We now put the separate engines into the order in which the formal proof uses them. This section does not rederive the negative-Laplace product or the Bromwich formula from the main article; it explains every transition from that exact representation to the all-orders statement.

23.1 Exact Lambert phase and exact normalized mass

Let $\lambda = \lambda(x)$ be the large positive solution of

$$\lambda 2^{-\lambda} = x, \qquad x \downarrow 0, \tag{247}$$

so

$$r = 2^{\lambda} = \frac{\lambda}{x}. \tag{248}$$

Write $\Lambda(r) = \log G(-r)$ for the negative log-transform. Define the dimensionless normalized saddle mass

$$\mathcal{S}(x) = \frac{F(x)\sqrt{2\pi\lambda}}{\exp(rx + \Lambda(r))}. \tag{249}$$

It is positive and satisfies the exact identity

$$\log F(x) = rx + \Lambda(r) - \frac{1}{2}\log(2\pi\lambda) + \log \mathcal{S}(x). \tag{250}$$

No asymptotic estimate has entered yet.

Let

$$\mathcal{M}(x) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda). \tag{251}$$

The exact negative-Laplace decomposition gives

$$rx + \Lambda(r) - \frac{1}{2}\log(2\pi\lambda) = \mathcal{M}(x) + E(r), \tag{252}$$

where the forward tail satisfies

$$0 \le E(2^\lambda) \le 4\exp(-2^\lambda). \tag{253}$$

Therefore

$$\log F(x) - \mathcal{M}(x) = \log \mathcal{S}(x) + E(2^\lambda). \tag{254}$$

The final term is flat relative to λ^{-1} by elementary exponential domination and will be reattached using theorem 14.3.

23.2 The central integral and the moving window

Bromwich inversion and the scaling $\theta = v/\sqrt{\lambda}$ give

$$\mathcal{S}(x) = \frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} \exp \Phi_\lambda(v/\sqrt{\lambda})\, \mathrm{d}v. \tag{255}$$

The exact phase separates as

$$\Phi_\lambda(v/\sqrt{\lambda}) = -\frac{v^2}{2} + \sum_{m=1}^{M} \varepsilon^m E_m(\lambda, v) + R_{M+1}(\lambda, v), \qquad \varepsilon = \lambda^{-1/2}, \tag{256}$$

inside an order-dependent central window

$$|v| \le A_N(\lambda) = \sqrt{32(N+1)\log\lambda}. \tag{257}$$

The all-orders Taylor modules prove, uniformly on this moving window, that the truncated perturbation tends to zero and that the analytic remainder is dominated by the first omitted power times a polynomial in v. The slow growth of A_N is crucial:

$$\lambda^{-1/2}A_N(\lambda)^d \longrightarrow 0 \qquad \text{for every fixed } d. \tag{258}$$

23.3 Finite exponential substitution

Fix the requested logarithmic order N. Choose the exponent and scalar-exponential truncation orders high enough that every omitted term is $O(\varepsilon^{2N}) = O(\lambda^{-N})$. Apply the exact finite-remainder identity (184) pointwise with

$$x = \varepsilon, \qquad E_m = E_m(\lambda, v). \tag{259}$$

This yields, on the central window,

$$\exp\left(\sum_{m<M} \varepsilon^m E_m\right) = \sum_{k<2N} \varepsilon^k a_k(\lambda, v) + O\big(\varepsilon^{2N} P_N(v)\big), \tag{260}$$

where P_N is a fixed-degree polynomial with bounded periodic coefficient families. The analytic remainder R_{M+1} is absorbed by a finite scalar exponential estimate because the whole perturbation is eventually bounded by one on the central window.

23.4 Gaussian contraction and the disappearance of half-orders

Multiply (260) by $\mathrm{e}^{-v^2/2}$ and integrate. Parity gives

$$\mathcal{G}[a_{2j+1}(\lambda, \cdot)] = 0. \tag{261}$$

For the even terms,

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \mathrm{e}^{-v^2/2} a_{2j}(\lambda, v)\, \mathrm{d}v = B_j(\lambda). \tag{262}$$

The central-window integral may be replaced by the whole-line integral because theorem 19.3 makes the complementary Gaussian-polynomial tail $O(\lambda^{-N})$.

The genuine saddle integrand outside the central window is controlled separately. The minor-arc estimates combine strict off-axis decay of the negative-Laplace product with far-tail polynomial bounds. They show that the noncentral part of (255) is also $O(\lambda^{-N})$ for every fixed N. Consequently

$$\mathcal{S}(x) = 1 + \sum_{j=1}^{N-1} \frac{B_j(\lambda)}{\lambda^j} + O(\lambda^{-N}). \tag{263}$$

This is the full mass expansion in the sense of theorem 14.1.

23.5 Logarithmic transfer and restoration of the flat tail

The ratio in (263) tends to 1 and is positive. Apply the formal logarithm recurrence and its finite analytic remainder theorem:

$$\log \mathcal{S}(x) = \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + O(\lambda^{-N}). \tag{264}$$

Now restore $E(2^\lambda)$ in (254). It is flat by (253), so theorem 14.3 leaves every coefficient unchanged.

Theorem 23.1 (Full exact-phase logarithmic expansion)**.** *For every $N \geq 1$, as $x \downarrow 0$,*

$$\boxed{\log F(x) = \mathcal{M}(x) + \sum_{j=1}^{N-1} \frac{A_j(\lambda(x))}{\lambda(x)^j} + O\big(\lambda(x)^{-N}\big).} \tag{265}$$

The coefficient functions A_j are bounded, smooth, and one-periodic; A_1 and A_2 are given by (242) *and* (245).

Proof. The preceding five subsections establish the expansion on the dyadic logarithmic coordinate. Use theorem 14.4 to transport it to the small-positive filter. The exact phase identity $\lambda(2^{-t})$ is retained throughout, so the coefficient families pull back without approximation. □

Because $\lambda(x)$ is comparable to $-\log x$, the remainder may be weakened to

$$O\big((-\log x)^{-N}\big), \tag{266}$$

but the coefficients still remain evaluated at the exact Lambert phase.

24 Why the exact phase cannot be casually re-expanded

The Lambert phase itself has an all-orders expansion in $T = -\log x$ and $\log T$. It is tempting to substitute that expansion into every $A_j(\lambda(x))$ and claim an ordinary nested-log series. The current formal theorem does not do so, for a good reason.

Let $\lambda_N(x)$ be a finite asymptotic approximation to $\lambda(x)$ and write

$$\Delta_N(x) = \lambda(x) - \lambda_N(x). \tag{267}$$

For a nonconstant periodic coefficient,

$$A_j(\lambda) - A_j(\lambda_N) = A_j'(\lambda_N)\Delta_N + \frac{1}{2}A_j''(\xi)\Delta_N^2. \tag{268}$$

To preserve an $O(\lambda^{-N})$ total remainder, the phase approximation must be accurate enough after multiplication by every prefactor λ^{-j} and after repeated Taylor composition through all coefficient families. A statement merely of the form $\Delta_N = O(\lambda^{-N})$ at one fixed order is not automatically stable under the entire triangular composition.

Warning 24.1 (The formalized boundary). The theorem proved in `FabiusFullAsymptoticExpansion.lean` is (265): every periodic coefficient is evaluated at the exact lower-Lambert phase. A separate branch expands the phase itself to all orders. What is not currently claimed is a single all-orders composition theorem that substitutes the phase series through every oscillatory coefficient while preserving the requested remainder. This is a missing formal composition layer, not a defect in the exact-phase expansion.

The exact-phase form is in any case mathematically preferable. It isolates the implicit saddle inversion once, avoids secular phase error, and makes phase-locked subsequences transparent. For any fixed $u \in \mathbb{R}$, the sequence

$$x_n = (n+u)2^{-(n+u)} \tag{269}$$

has exact phase $\lambda(x_n) = n + u$, so every periodic coefficient is literally constant along it:

$$A_j(\lambda(x_n)) = A_j(u). \tag{270}$$

This is the cleanest way to separate algebraic decay from periodic oscillation.

A Executable recurrences in mathematical pseudocode

The following algorithms are not substitutes for the proofs above. They are compact implementations of the finite algebra that appears repeatedly in the formal development. All arrays are zero-indexed.

Algorithm A.1 (Weighted path sum)**.** The dynamic program below computes $P_w(n)$ without enumerating compositions. It is the triangular recurrence proved equivalent to the composition sum in theorem 10.2.

```
function PathSums(N, weight):
    P[0] := 1
    for n := 1 to N:
        P[n] := 0
        for k := 0 to n - 1:
            P[n] := P[n] + P[k] * weight(k, n)
    return P

function FabiusInverseDyadic(N):
    weight(k, n) := 1 / ((2^n - 1) * factorial(n - k + 1))
    a := PathSums(N, weight)
    for n := 0 to N:
        value[n] := 2^(-binomial(n, 2)) * a[n]
    return value
```

Algorithm A.2 (Formal exponential coefficients). Given $E[1], \ldots, E[N]$ in a commutative $\mathbb{Q}$-algebra, this returns the first N coefficients of $\exp(\sum_{j\geq 1} E[j]u^j)$.

```
function ExpCoeff(E, N):
    a[0] := 1
    for n := 0 to N - 1:
        s := 0
        for j := 0 to n:
            s := s + (j + 1) * E[j + 1] * a[n - j]
        a[n + 1] := s / (n + 1)
    return a
```

Algorithm A.3 (Formal logarithm coefficients). The input is a unit series $1 + \sum_{j\geq 1} a[j]u^j$. The output satisfies $\log(1 + \sum a[j]u^j) = \sum b[j]u^j$.

```
function LogCoeff(a, N):
    b[0] := 0
    for n := 0 to N - 1:
        s := 0
        for j := 0 to n - 1:
            s := s + (n - j) * b[n - j] * a[j + 1]
        b[n + 1] := a[n + 1] - s / (n + 1)
    return b
```

Algorithm A.4 (Normalized Gaussian contraction). The moment table can be built once to the degree of the input polynomial.

```
function GaussianMoments(D):
    mu[0] := 1
    if D >= 1: mu[1] := 0
    for n := 0 to D - 2:
        mu[n + 2] := (n + 1) * mu[n]
    return mu

function GaussianContract(coeff, D):
    mu := GaussianMoments(D)
    result := 0
    for k := 0 to D:
```

```
        result := result + coeff[k] * mu[k]
    return result
```

Algorithm A.5 (Saddle coefficient pipeline)**.** This is the finite algebra at fixed logarithmic order N. The analytic proof is what justifies applying it to the central integral with a uniform remainder.

```
function SaddleCoefficients(N, phase):
    # 1. Build bounded exponent jets Z[0 .. 2N - 1].
    Z := ClosedJets(2*N - 1, phase)

    # 2. Build exponent polynomials E[1 .. 2N].
    for m := 1 to 2*N:
        E[m] := i^m * (Z[m - 1] * X^m / factorial(m)
                       + (-1)^(m + 1) * X^(m + 2) / (m + 2))

    # 3. Exponentiate formally in epsilon.
    a := ExpCoeff(E, 2*N)

    # 4. Contract even epsilon orders against the normalized Gaussian.
    B[0] := 1
    for j := 1 to N:
        B[j] := GaussianContract(coefficients(a[2*j]), degree(a[2*j]))

    # 5. Convert mass coefficients to logarithmic coefficients.
    A := LogCoeff(B, N)
    return (B, A)
```

B Lean declaration and module concordance

File names below are relative to

`Analysis/FabiusFunction/Lean/FabiusFunction/`.

The table lists the principal declarations, not every supporting lemma.

Module	Content explained in this companion
`OriginalCharacterization.lean`	`IsOriginalFabius.mk_of_derivative_law`: the support/mean-value argument showing that scale positivity follows from the other original-characterization assumptions; section 1.3.
`FabiusInverse.lean`	`fabiusOrderIso`, `fabiusInv`, `fabiusReal_fabiusInv`, `fabiusInv_fabiusReal`, `continuous_fabiusInv`, `strictMonoOn_fabiusInv`, `fabiusInv_one_sub`, `fabiusInv_fabiusDyadicUnit`, `fabiusInv_hasDerivAt`, `deriv_fabiusInv`, `deriv_fabiusInv_eq_inv_two_mul_rvachevUp`, `deriv_fabiusInv_pos`, `fabiusInv_contDiffOn_Ioo`, the transported root-free bounds, `mul_lt_fabiusInv`, `tendsto_fabiusInv_div_atTop`, and reflected companions; sections 2 and 3.
`ElementaryFunction.lean`	The inductive predicate `IsElementary`, composition closure, derived elementary operations, analytic loci, the finite-bad-value composition theorem, and the dense-open analytic-locus theorem; section 4.

Module	Content explained in this companion
`NotElementary.lean`	No elementary function agrees with a bounded Fabius function, up, or the signed extension on a set with nonempty interior; section 5.
`ProbabilityRepresentation.lean`	The product sample space, uniformly convergent coordinate sum, pushforward law, head–tail split, coordinate reflection, and identification of the CDF with F; section 6.
`ProbabilityLaplaceMoments.lean`	Almost-sure support, restriction to $[0, 1]$, survival function, the compact-support Fubini identity, raw and endpoint moment identifications, zero-tilt normalization, and positivity; sections 7 and 8.
`LaplaceMoments.lean`	`tiltedSurvivalMoment`, `fabiusLaplaceMoment`, the differential recurrence, iterated derivatives, smoothness, and evaluation at zero tilt; section 8.
`StepMeasureBridge.lean`	Almost-everywhere replacement of half-endpoint indicators, exact overlap integrals, atomic measure comparisons, the shrunken/enlarged interval sandwich, portmanteau, and convergence of every interval integral; section 9.
`FabiusInverseDyadicClosedForm.lean`	Compositions, path weights, last-edge decomposition, generic triangular solver, uniqueness, and the exact composition formula for $F(2^{-n})$; sections 10 and 11.
`FabiusDiscreteLimitToeplitz.lean`	Corrected prefix length, exact q-binomial row mass, total-variation bound 16, the generic finite-row Toeplitz theorem, and the specialization $q(n, j) = 2n - j$; sections 12 and 13.
`SaddleExpansionAlgebra.lean`	`expCoeff`, its differential equation, uniqueness, naturality, rescaling, addition, parity, and `HasAsymptoticExpansion`; sections 14 and 15.
`SaddleLogExpansionAlgebra.lean`	`logCoeff`, the identity $AB' = A'$, low-order formulas, uniqueness, naturality, rescaling, and parity; section 16.
`SaddleExpansionFiniteRemainder.lean`	Finite exponent substitution, equality of coefficients below the truncation order, divisibility of the defect by X^L, and the evaluated exact remainder; section 17.
`SaddleExpansionSmallArgument.lean`	Exact transfer of full expansions between $t \to \infty$ and $x = 2^{-t} \downarrow 0$; section 14.
`SaddleExpansionFlat.lean`	Zero-coefficient expansion of all-orders-flat errors and invariance of every coefficient under addition or subtraction of such an error; sections 14 and 23.
`GaussianPolynomialContraction.lean`	Gaussian integrability, odd moments, integration-by-parts recurrence, normalized moments, linear polynomial contraction, and the whole-line integral identity; section 18.
`GaussianPolynomialTail.lean`	The factorial coefficient weight and explicit complementary-interval tail bound; section 19.
`GaussianPolynomialTailAllOrders.lean`	The radius $\sqrt{32(N+1)\log b}$ and arbitrary algebraic tail order for bounded polynomial families; section 19.
`FabiusSaddleJetClosedForm.lean`	Closed formula for the periodic and bounded exponent jets, including harmonic constants and the shifted falling-factorial coefficients; section 20.
`FabiusSaddleJetStirling.lean`	Identification $c(n, m) = s(n+1, m+1)$ with signed Stirling numbers of the first kind; section 20.
`FabiusSaddleExponentClosedForm.lean`	Cancellation of factorials and the extra power of i, yielding the two-monomial formula (217); section 20.
`FabiusSaddleLeadingCoefficient.lean`	The unique newest-jet monomial and coefficient $(-1)^j/(2^j j!)$ in the jth mass coefficient; section 21.
`FabiusSecondSaddleCorrection.lean`	Low-order formal exponential identities, explicit order-four polynomial, Gaussian contraction, closed B_2, and closed A_2; section 22.

Module	Content explained in this companion
`FabiusSecondSaddleExpansion.lean`	The explicit expansion through A_2/λ^2 with $O(\lambda^{-3})$ and the weakened $O((-\log x)^{-3})$ form; sections 22 and 23.
`FabiusSaddleMassAllOrders.lean`	Finite exponential substitution, bounded jet families, central reference polynomial, Gaussian whole-line contraction, and all-orders normalized saddle mass; section 23.
`FabiusFullAsymptoticExpansion.lean`	Transfer from complex kernel mass to the real saddle ratio, logarithmic coefficient transfer, addition of the flat forward tail, exact-phase full expansion, and small-argument transport; sections 23 and 24.

C Audit matrix against the main article

Topic	Status before this companion	Added here
Original scale positivity	Assumed in the source statement	Mean-value proof that the dilation parameter is automatically positive before its exact value $k = 2$ is derived.
Total inverse I	Absent	Global clamped construction, order calculus, reflection, exact dyadic inverse values, interior calculus consequences, endpoint steepness, and non-Hölder corollaries.
Elementary function class	Absent	Exact inductive closure, totalization convention, finite-bad-value composition, and dense-open analytic locus.
Non-elementarity	Absent	No local agreement on any set with nonempty interior for F, up, and the signed extension.
Product probability law	Headline proof only	Uniform convergence and measurability, head–tail pushforward identity, coordinate reflection, and global support bookkeeping.
Probability–Laplace bridge	Absent	Survival-function Fubini identity, identification of raw, endpoint, and tilted moments, complete monotonicity, and half-moment interpretation.
Atomic-to-step bridge	Compressed	Half-endpoint null-set conversion, exact cell overlap, shrunken/enlarged interval sandwich, portmanteau, and moving-boundary squeeze.
Weighted path solver	Explicitly omitted	Generic semiring theorem, uniqueness, edge-count decomposition, Fabius specialization, and worked values.
Finite-row Toeplitz theorem	Explicitly omitted	Abstract normed-ring proof, exact Fabius row mass, uniform variation bound, tail-index condition, and corrected prefix rounding.
Parameter-dependent full expansions	Only summarized	Exact API, coefficient boundedness, closure, flat errors, and coordinate transfer.
Formal exponential/logarithm	Explicitly omitted	Recurrences, formal differential equations, uniqueness, low orders, naturality, rescaling, addition, and parity.
Finite formal-to-analytic remainder	Absent	Exact divisibility by the first omitted power and evaluated polynomial remainder.
Gaussian contraction/tails	Explicitly omitted	Moment recurrence, contraction functional, factorial coefficient weight, explicit tail, and all-orders radius.

Topic	Status before this companion	Added here
Closed saddle jets	Stated, lightly motivated	Operator-product solution, harmonic constants, shifted signed Stirling identification, and first four bounded jets.
Second saddle correction	Not yet in pinned snapshot	Complete E_1–E_4 expansion, order-four polynomial, Gaussian contraction, B_2, and closed A_2.
Full saddle assembly	Proof architecture only	Exact sequence of finite remainder, central window, contraction, Gaussian tail, minor arc, log transfer, flat-tail restoration, and filter transport.
Exact phase boundary	Warning only	Quantified explanation of why all-orders substitution through oscillatory coefficients needs a separate composition theorem.

The following substantial topics are intentionally not repeated because the main article already treats them at proof level: construction and uniqueness of F and up; the original Rvachev characterization apart from the redundancy isolated in section 1.3; the infinite sinc product and Fourier decay; support, positivity, shape, and nowhere analyticity; exact moment recurrences and denominator arithmetic; dyadic rational evaluation, q-Pochhammer and q-binomial formulas; Thue–Morse interpolation and iterated difference identities; Legendre expansions and least-squares approximation; and the computational API. Their use as input above is always signposted.

D Notation crosswalk

This companion	Main-article notation	Principal Lean declaration
$I(y)$	inverse of bounded F	`fabiusInv`
μ	law of the weighted uniform sum	`weightedSumDistribution`
$J_k(s)$	tilted survival moment	`tiltedSurvivalMoment`
$M_k(s)$	tilted probability/Fabius moment	`fabiusLaplaceMoment`
d_n	exact half moment	`halfMoment`
$P_w(n)$	weighted composition/path sum	generic path-sum definition in `FabiusInverseDyadicClosedForm.lean`
σ	abstract small scale	first argument of `HasAsymptoticExpansion`
a_n	formal exponential coefficient	`SaddleExpansion.expCoeff`
b_n	formal logarithm coefficient	`SaddleExpansion.logCoeff`
$\mathcal{G}[p]$	normalized Gaussian contraction	`SaddleExpansion.gaussianPolynomialContraction`
$Z_n(t)$	bounded exponent jet $\widetilde{\mathcal{Z}}_n$	`negativeLaplaceBoundedExponentJet`
$B_j(t)$	normalized saddle mass coefficient	`fabiusSaddleMassCoefficient`
$A_j(t)$	logarithmic saddle coefficient	`fabiusSaddleLogCoefficient`
$\lambda(x)$	exact lower-Lambert phase	`fabiusLambertPhase`
$\mathcal{M}(x)$	corrected sharp Lambert main	`fabiusSharpLambertMain`

This companion	Main-article notation	Principal Lean declaration
$\mathcal{S}(x)$	normalized real saddle mass/ratio	`fabiusSaddleRatio` and the kernel-mass bridge

E A compact dependency diagram

The proof dependencies of the material unique to this companion can be read from the following diagram. An arrow means that the target uses the source as a mathematical input, not merely as a Lean import.

$$
\begin{aligned}
&\text{support + interior positivity + mean-value theorem + derivative law} \longrightarrow k > 0,\\
&\text{strict monotonicity + surjectivity + flatness} \longrightarrow \text{total inverse} \longrightarrow \text{endpoint steepness},\\
&\text{nowhere analyticity + dense elementary analytic locus} \longrightarrow \text{strong non-elementarity},\\
&\text{product law} \longrightarrow \text{survival identity} \longrightarrow \text{Laplace/raw/endpoint moment bridge},\\
&\text{weak convergence + null boundaries + cell geometry} \longrightarrow \text{step-integral convergence},\\
&\text{triangular recurrence} \longrightarrow \text{weighted paths} \longrightarrow \text{composition formula},\\
&q\text{-binomial row mass + variation + tail indices} \longrightarrow \text{discrete Toeplitz limit},\\
&\text{closed jets} \longrightarrow \text{exponent coefficients} \longrightarrow \text{formal exponential}\\
&\qquad\qquad\qquad \longrightarrow \text{Gaussian contraction} \longrightarrow \text{mass coefficients},\\
&\text{mass coefficients} \longrightarrow \text{formal logarithm} \longrightarrow \text{log coefficients},\\
&\text{finite remainders + central Taylor + Gaussian tails + minor arcs}\\
&\qquad\qquad\qquad \longrightarrow \text{all-orders saddle mass},\\
&\text{all-orders mass + exact reduction + flat tail + coordinate transfer}\\
&\qquad\qquad\qquad \longrightarrow \text{full exact-phase expansion}.
\end{aligned}
$$

References

[1] J. Fabius, A probabilistic example of a nowhere analytic C^∞-function, *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* **5** (1966), 173–174.

[2] J. Arias de Reyna, On the Fabius function, *The American Mathematical Monthly* **89** (1982), 291–298.

[3] J. Arias de Reyna, An infinitely differentiable function with compact support: Definition and properties, arXiv:1702.05442.

[4] J. Arias de Reyna, The asymptotic behavior of the Fabius function, arXiv:1702.06487, version 3, 2018.

[5] V. Reshetnikov, *The Fabius Function and Rvachev's Up-Function: Exact Arithmetic, Probability, Fourier Analysis, and Corrected Sharp Asymptotics*, `ProveIt`, `Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up/`.

[6] V. Reshetnikov, `ProveIt`: formal development of the Fabius function and Rvachev's up-function, GitHub repository, snapshot `5e80805b1ee92697723949d125aa0d6dbf32f538`, 25 August 2026, `https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction`.